T3 general symbolic | UV

General scalar and Yukawa interactions; expressions reproduced in native InputForm
Matching and physical Matchete/RGBeta coupling mapping NOT CERTIFIED

MODEL A_+0

Extra quartics: 5; extra bilinears: 1; extra Yukawa: y1Conj

COMPLETE BSM LAGRANGIAN

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Bar[Field[NewScalar1, Scalar, {}, {}]]^2*Coupling[lambdaT3ExtraM000200I1, {}, 0] + Bar[Field[NewScalar1, Scalar, {},
{}]]^4*Coupling[lambdaT3ExtraQ000400I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]^2*CG[T3ExtraCGQ110200I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1,
SU2L[fund]]}]*Coupling[lambdaT3ExtraQ110200I1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] +
Bar[Coupling[lambdaT3ExtraQ021010I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3ExtraCGQ021010I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]
+ Bar[Field[NewScalar1, Scalar, {}, {}]]^3*Coupling[lambdaT3ExtraQ001300I1, {}, 0]*Field[NewScalar1, Scalar, {}, {}] -
Bar[Field[NewScalar1, Scalar, {}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {}, {}] + Bar[Field[H, Scalar,
{Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1,
SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG,
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H,
Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {},
{}] + Bar[Coupling[lambdaT3ExtraM000200I1, {}, 0]]*Field[NewScalar1, Scalar, {}, {}]^2 + (Bar[Field[NewScalar1, Scalar,
 {}, {}]]^2*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {}, {}]^2)/2 + Bar[Coupling[lambdaT3ExtraQ110200I1,
 {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[T3ExtraCGQ110200I1, {Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, SU2L[fund]]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {}, {}]^2 +
Bar[Coupling[lambdaT3ExtraQ001300I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*Field[NewScalar1, Scalar, {}, {}]^3
 + Bar[Coupling[lambdaT3ExtraQ000400I1, {}, 0]]*Field[NewScalar1, Scalar, {}, {}]^4 + Bar[Field[NewScalar1, Scalar, {},
 {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {}, {Index[d$$1, Lorentz]}] + Bar[Coupling[lambdaT3, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]],
Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] - Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ021010I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ021010I1, {}, 0]*Field[NewScalar1, Scalar, {}, {}]*Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + Bar[Field[NewScalar1, Scalar,
{}, {}]]^2*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3ExtraCGQ000211I1, {Index[d$$2, T3BSMd3],
Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3ExtraQ000211I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
(Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
 T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar1, Scalar, {}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambda12Inv1, {},
0]*Field[NewScalar1, Scalar, {}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
Bar[Coupling[lambdaT3ExtraQ000211I1, {}, 0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000211I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar1, Scalar, {},
{}]^2*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv2,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv2, {},
0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 -
Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[T3Y1ConjCG, {Index[d$$1, SU2L[fund]],
Index[d$$1, T3BSMd2]}]*Coupling[y1Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion,
{Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Index[d$$1,
T3BSMd2]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar, {},
{}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]],
Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] +
I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG],
{Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
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DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[Bar[T3Y1CG],
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {}]] + Bar[Coupling[y1Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y1ConjCG],
 {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]]}]*Field[NewScalar1, Scalar, {}, {}]*NCM[Transp[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
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UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(59*gY^3)/6

beta_g2 =
-1/2*g2^3

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (2*Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + 2*Matrix[y1Conj,
Trans[Bar[y1Conj]], ye][gen[$i], gen[$j]] + 3*Matrix[y2, Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 6*Matrix[ye,
Trans[Bar[ye]], ye][gen[$i], gen[$j]])/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + 2*Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]]*Matrix[y1][gen[$i], heavy[$j]] +
4*Tr[y1 . Trans[Bar[y1Conj]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + (4*Matrix[y1, Trans[Bar[y1]], y1][gen[$i],
heavy[$j]] + 10*Matrix[y1, Trans[Bar[y1Conj]], y1Conj][gen[$i], heavy[$j]] + 10*Matrix[y1Conj, Trans[Bar[y1Conj]],
y1][gen[$i], heavy[$j]] + 3*Matrix[y2, Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]],
y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + Matrix[y1Conj,
Trans[Bar[y1Conj]], y2][gen[$i], heavy[$j]] + 3*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[ye,
Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/2

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (3*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + 2*(Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]] +
Matrix[Trans[y1Conj], Bar[y1Conj], MF][heavy[$i], heavy[$j]]))/4

beta_mS1Sq =
6*lambda12*mS2Sq + 12*lambdaT3GeneralQ001300I1*Bar[muT3GeneralS1Sq] + muT3GeneralS1Sq*(12*Bar[lambdaT3GeneralQ001300I1]
 + 8*Tr[y1 . Trans[Bar[y1Conj]]]) + 8*Bar[muT3GeneralS1Sq]*Tr[y1Conj . Trans[Bar[y1]]] + mS1Sq*(4*lambdaS1 + 4*Tr[y1 .
Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - 8*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]] - 8*Tr[MF .
Trans[Bar[MF]] . Trans[y1Conj] . Bar[y1Conj]]

beta_mS2Sq =
2*lambda12*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + 4*muT3GeneralS1Sq*Bar[lambdaT3GeneralQ000211I1] +
4*lambdaT3GeneralQ000211I1*Bar[muT3GeneralS1Sq] + mS2Sq*(8*lambdaS2 + 4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]]) -
4*Tr[MF . Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 2*lambdaH1^2 + 6*lambdaH2^2 +
 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] +
8*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
6*lambda12^2 + 4*lambdaH1^2 + 10*lambdaS1^2 + 12*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1] +
288*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1] + 8*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] +
 lambdaT3GeneralQ001300I1*(108*Bar[lambdaT3GeneralQ001300I1] + 24*Tr[y1 . Trans[Bar[y1Conj]]]) +
24*Bar[lambdaT3GeneralQ001300I1]*Tr[y1Conj . Trans[Bar[y1]]] + lambdaS1*(8*Tr[y1 . Trans[Bar[y1]]] + 8*Tr[y1Conj .
Trans[Bar[y1Conj]]]) - 8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]] - 16*Tr[y1 . Trans[Bar[y1]] . y1Conj .
Trans[Bar[y1Conj]]] - 16*Tr[y1 . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1]]] - 8*Tr[y1Conj . Trans[Bar[y1Conj]] .
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y1Conj . Trans[Bar[y1Conj]]]

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH2^2 + gY^2*(8*g2^2 - 12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 +
16*lambdaS2Adj^2 + 8*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1] + lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 .
Trans[Bar[y2]]]) - (8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaH1 =
(-9*g2^2*lambdaH1)/2 - (3*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambda12*lambdaH2 +
4*lambdaH1*lambdaS1 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] +
12*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ110200I1] +
lambdaT3GeneralQ110200I1*(4*(3*Bar[lambdaT3GeneralQ001300I1] + 4*Bar[lambdaT3GeneralQ110200I1]) + 8*Tr[y1 .
Trans[Bar[y1Conj]]]) + 8*Bar[lambdaT3GeneralQ110200I1]*Tr[y1Conj . Trans[Bar[y1]]] + lambdaH1*(4*Tr[y1 .
Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]]) - 4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]] - 4*Tr[y1Conj . Trans[Bar[y1Conj]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + 2*lambda12*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2
 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ000211I1] + 4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] +
4*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 4*lambda12*lambdaS1 + 8*lambda12*lambdaS2 +
4*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ000211I1] +
4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + lambdaT3GeneralQ000211I1*(4*(4*Bar[lambdaT3GeneralQ000211I1]
 + 3*Bar[lambdaT3GeneralQ001300I1]) + 8*Tr[y1 . Trans[Bar[y1Conj]]]) + 8*Bar[lambdaT3GeneralQ000211I1]*Tr[y1Conj .
Trans[Bar[y1]]] + lambda12*(4*lambdaS2Adj + 4*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]] + 2*Tr[y2 .
Trans[Bar[y2]]]) - 4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]] - 4*Tr[y1Conj . Trans[Bar[y1Conj]] . y2 .
Trans[Bar[y2]]]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 4*Bar[lambdaT3GeneralQ021010I1]*(Bar[lambdaT3GeneralQ000211I1] + 2*Bar[lambdaT3GeneralQ110200I1])
 + 4*Bar[lambdaT3GeneralQ021010I1]*Tr[y1 . Trans[Bar[y1Conj]]] + lambdaT3*(4*lambdaH2Adj + 2*Tr[y1 . Trans[Bar[y1]]] +
2*Tr[y1Conj . Trans[Bar[y1Conj]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
gY^2*(4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*lambdaH2Adj +
2*lambdaH2Adj*lambdaS2 + (8*lambdaT3*Bar[lambdaT3])/3 + (8*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1])/3 +
lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
6*Tr[yu . Trans[Bar[yu]]]) + (4*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]])/3

beta_lambdaS2Adj =
-2*g2^4 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj +
20*lambdaS2Adj^2 + 4*lambdaS2Adj*Tr[y2 . Trans[Bar[y2]]] - (2*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_y1Conj =
4*Tr[y1Conj . Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y1Conj][gen[$i], heavy[$j]])/2 -
(3*gY^2*Matrix[y1Conj][gen[$i], heavy[$j]])/2 + 2*Tr[y1 . Trans[Bar[y1]]]*Matrix[y1Conj][gen[$i], heavy[$j]] +
2*Tr[y1Conj . Trans[Bar[y1Conj]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + (10*Matrix[y1, Trans[Bar[y1]], y1Conj][gen[$i],
heavy[$j]] + 10*Matrix[y1Conj, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 4*Matrix[y1Conj, Trans[Bar[y1Conj]],
y1Conj][gen[$i], heavy[$j]] + 3*Matrix[y2, Trans[Bar[y2]], y1Conj][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]],
y1Conj][gen[$i], heavy[$j]])/4

beta_lambdaT3GeneralQ000211I1 =
-12*g2^2*lambdaT3GeneralQ000211I1 - 6*gY^2*lambdaT3GeneralQ000211I1 + 2*lambdaS1*lambdaT3GeneralQ000211I1 +
8*lambdaS2*lambdaT3GeneralQ000211I1 + 4*lambdaS2Adj*lambdaT3GeneralQ000211I1 + 4*lambdaH2*lambdaT3GeneralQ110200I1 +
24*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000211I1] + 4*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] +
lambda12*(8*lambdaT3GeneralQ000211I1 + 6*lambdaT3GeneralQ001300I1 + 4*Tr[y1Conj . Trans[Bar[y1]]]) +
lambdaT3GeneralQ000211I1*(4*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]]) -
4*Tr[y1Conj . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]]

beta_lambdaT3GeneralQ000400I1 =
3*lambdaT3GeneralQ000211I1^2 + 12*lambdaS1*lambdaT3GeneralQ000400I1 + 9*lambdaT3GeneralQ001300I1^2 +
2*lambdaT3GeneralQ110200I1^2 + 4*lambdaT3GeneralQ001300I1*Tr[y1Conj . Trans[Bar[y1]]] +
lambdaT3GeneralQ000400I1*(8*Tr[y1 . Trans[Bar[y1]]] + 8*Tr[y1Conj . Trans[Bar[y1Conj]]]) - 4*Tr[y1Conj . Trans[Bar[y1]]
 . y1Conj . Trans[Bar[y1]]]

beta_lambdaT3GeneralQ001300I1 =
6*lambda12*lambdaT3GeneralQ000211I1 + 4*lambdaH1*lambdaT3GeneralQ110200I1 +
lambdaT3GeneralQ000400I1*(72*Bar[lambdaT3GeneralQ001300I1] + 16*Tr[y1 . Trans[Bar[y1Conj]]]) +
lambdaS1*(18*lambdaT3GeneralQ001300I1 + 4*Tr[y1Conj . Trans[Bar[y1]]]) + lambdaT3GeneralQ001300I1*(8*Tr[y1 .
Trans[Bar[y1]]] + 8*Tr[y1Conj . Trans[Bar[y1Conj]]]) - 8*Tr[y1 . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1]]] -
8*Tr[y1Conj . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1Conj]]]

beta_lambdaT3GeneralQ021010I1 =
```

```
(-21*g2^2*lambdaT3GeneralQ021010I1)/2 - (9*gY^2*lambdaT3GeneralQ021010I1)/2 + 2*lambda12*lambdaT3GeneralQ021010I1 +
2*lambdaH*lambdaT3GeneralQ021010I1 + 4*lambdaH1*lambdaT3GeneralQ021010I1 + 4*lambdaH2*lambdaT3GeneralQ021010I1 +
4*lambdaH2Adj*lambdaT3GeneralQ021010I1 + 4*Bar[lambdaT3]*(Bar[lambdaT3GeneralQ000211I1] +
2*Bar[lambdaT3GeneralQ110200I1]) + 4*Bar[lambdaT3]*Tr[y1 . Trans[Bar[y1Conj]]] + lambdaT3GeneralQ021010I1*(2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye .
Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110200I1 =
6*lambdaH2*lambdaT3GeneralQ000211I1 - (9*g2^2*lambdaT3GeneralQ110200I1)/2 - (3*gY^2*lambdaT3GeneralQ110200I1)/2 +
6*lambdaH*lambdaT3GeneralQ110200I1 + 2*lambdaS1*lambdaT3GeneralQ110200I1 +
12*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] + 24*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ110200I1] +
lambdaH1*(6*lambdaT3GeneralQ001300I1 + 8*lambdaT3GeneralQ110200I1 + 4*Tr[y1Conj . Trans[Bar[y1]]]) +
lambdaT3GeneralQ110200I1*(4*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y1Conj . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_muT3GeneralS1Sq =
6*lambdaT3GeneralQ000211I1*mS2Sq + 2*lambdaS1*muT3GeneralS1Sq + 24*lambdaT3GeneralQ000400I1*Bar[muT3GeneralS1Sq] +
mS1Sq*(6*lambdaT3GeneralQ001300I1 + 4*Tr[y1Conj . Trans[Bar[y1]]]) + muT3GeneralS1Sq*(4*Tr[y1 . Trans[Bar[y1]]] +
4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - 8*Tr[MF . Trans[Bar[MF]] . Trans[y1Conj] . Bar[y1]]
```

MODEL A_-2

Extra quartics: 5; extra bilinears: 1; extra Yukawa: y2Conj

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGM000002I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraM000002I1, {}, 0] +
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000004I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ000004I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3ExtraCGQ020101I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ020101I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGQ110002I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ110002I1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] +
Bar[Field[NewScalar1, Scalar, {}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2,
 Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3ExtraCGQ001102I1, {Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ001102I1, {}, 0]*Field[NewScalar1, Scalar, {}, {}] - Bar[Field[NewScalar1, Scalar,
{}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2,
SU2L[fund]]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {}, {}] +
(Bar[Field[NewScalar1, Scalar, {}, {}]]^2*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {}, {}]^2)/2 +
Bar[Field[NewScalar1, Scalar, {}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] - Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] +
Bar[Coupling[lambdaT3ExtraQ020101I1, {}, 0]]*CG[Bar[T3ExtraCGQ020101I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2,
 SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {}, {}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] +
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar1, Scalar, {},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$2, T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraM000002I1, {}, 0]]*CG[T3ExtraCGM000002I1, {Index[d$$1,
T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ110002I1, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]]*CG[T3ExtraCGQ110002I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3],
 Index[d$$2, T3BSMd3]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ001102I1, {},
0]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[T3ExtraCGQ001102I1, {Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Field[NewScalar1, Scalar, {}, {}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*CG[T3ExtraCGQ000013I1,
{Index[d$$4, T3BSMd3], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3,
```

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T3BSMd3]}]*Coupling[lambdaT3ExtraQ000013I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
(Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv2, {}, 0]*Field[NewScalar2,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 +
Bar[Coupling[lambdaT3ExtraQ000013I1, {}, 0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000013I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ000004I1, {},
0]]*CG[T3ExtraCGQ000004I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] -
Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {}]] + CG[T3Y2ConjCG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*Coupling[y2Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]],
Index[d$$1, T3BSMd2]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar, {},
{}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]],
Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] +
I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{Index[d$$1, Lorentz]}]] + Bar[Coupling[y2Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3Y2ConjCG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2],
Index[d$$1, T3BSMd3]}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y2,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2],
Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]],
 Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1,
Scalar, {}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]]}]*NCM[Transp[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(55*gY^3)/6

beta_g2 =
-1/2*g2^3

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (2*Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + 3*(Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + Matrix[y2Conj, Trans[Bar[y2Conj]], ye][gen[$i], gen[$j]] + 2*Matrix[ye,
Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + 2*Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (4*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2Conj, Trans[Bar[y2Conj]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye,
Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + Tr[y2Conj . Trans[Bar[y2Conj]]]*Matrix[y2][gen[$i], heavy[$j]] +
2*Tr[y2 . Trans[Bar[y2Conj]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + (6*Matrix[y1, Trans[Bar[y1]], y2][gen[$i],
heavy[$j]] + 18*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 29*Matrix[y2, Trans[Bar[y2Conj]], y2Conj][gen[$i],
 heavy[$j]] + 29*Matrix[y2Conj, Trans[Bar[y2Conj]], y2][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]], y2][gen[$i],
```

```
 heavy[$j]])/12

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (3*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + 3*Matrix[MF, Trans[Bar[y2Conj]], y2Conj][heavy[$i], heavy[$j]] +
2*Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]])/4

beta_mS1Sq =
-6*gY^2*mS1Sq + 6*lambda12*mS2Sq + 12*muT3GeneralS2Sq*Bar[lambdaT3GeneralQ001102I1] +
12*lambdaT3GeneralQ001102I1*Bar[muT3GeneralS2Sq] + mS1Sq*(4*lambdaS1 + 4*Tr[y1 . Trans[Bar[y1]]]) - 8*Tr[MF .
Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
2*lambda12*mS1Sq - 12*g2^2*mS2Sq + 20*lambdaT3GeneralQ000013I1*Bar[muT3GeneralS2Sq] + 4*Bar[muT3GeneralS2Sq]*Tr[y2 .
Trans[Bar[y2Conj]]] + muT3GeneralS2Sq*(20*Bar[lambdaT3GeneralQ000013I1] + 4*Tr[y2Conj . Trans[Bar[y2]]]) +
mS2Sq*(8*lambdaS2 + 4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 4*Tr[MF .
Trans[Bar[y2]] . y2 . Trans[Bar[MF]]] - 4*Tr[MF . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 2*lambdaH1^2 + 6*lambdaH2^2 +
 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] +
24*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
12*gY^4 + 6*lambda12^2 + 4*lambdaH1^2 - 12*gY^2*lambdaS1 + 10*lambdaS1^2 +
24*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1] + 8*lambdaS1*Tr[y1 . Trans[Bar[y1]]] - 8*Tr[y1 .
Trans[Bar[y1]] . y1 . Trans[Bar[y1]]]

beta_lambdaS2 =
22*g2^4 + 2*lambda12^2 + 4*lambdaH2^2 - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 16*lambdaS2Adj^2 +
(832*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 +
(8*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1])/3 +
(16*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3 + (32*Bar[lambdaT3GeneralQ000013I1]*Tr[y2 .
Trans[Bar[y2Conj]]])/3 + lambdaT3GeneralQ000013I1*((352*Bar[lambdaT3GeneralQ000013I1])/3 + (32*Tr[y2Conj .
Trans[Bar[y2]]])/3) + lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
(8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3 - (248*Tr[y2 . Trans[Bar[y2]] . y2Conj . Trans[Bar[y2Conj]]])/45 -
(248*Tr[y2 . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2]]])/45 - (8*Tr[y2Conj . Trans[Bar[y2Conj]] . y2Conj .
Trans[Bar[y2Conj]]])/3

beta_lambdaH1 =
3*gY^4 - (9*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambda12*lambdaH2 +
4*lambdaH1*lambdaS1 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ001102I1] +
12*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 12*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ110002I1] +
 lambdaH1*(4*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]]) - 4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + 2*lambda12*lambdaH1 - (33*g2^2*lambdaH2)/2 - (3*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 20*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ110002I1] +
4*Bar[lambdaT3GeneralQ110002I1]*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ110002I1*(4*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ110002I1]) + 4*Tr[y2Conj .
Trans[Bar[y2]]]) + lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]] - 2*Tr[y2Conj . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 4*lambda12*lambdaS1 + 8*lambda12*lambdaS2 +
4*lambdaT3*Bar[lambdaT3] + 20*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ001102I1] +
4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 4*Bar[lambdaT3GeneralQ001102I1]*Tr[y2 . Trans[Bar[y2Conj]]]
+ lambdaT3GeneralQ001102I1*(4*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ001102I1]) + 4*Tr[y2Conj .
Trans[Bar[y2]]]) + lambda12*(4*lambdaS2Adj + 4*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj .
Trans[Bar[y2Conj]]]) - 4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]] - 4*Tr[y1 . Trans[Bar[y1]] . y2Conj .
Trans[Bar[y2Conj]]]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ020101I1] +
8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ020101I1] + 2*Bar[lambdaT3GeneralQ020101I1]*Tr[y2 . Trans[Bar[y2Conj]]]
+ lambdaT3*(4*lambdaH2Adj + 2*Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + Tr[y2Conj . Trans[Bar[y2Conj]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
(-33*g2^2*lambdaH2Adj)/2 - (3*gY^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*lambdaH2Adj +
2*lambdaH2Adj*lambdaS2 + (8*lambdaT3*Bar[lambdaT3])/3 - (8*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1])/3 +
lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]]
 + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (4*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]])/3 -
(4*Tr[y2Conj . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]])/3
```

```
beta_lambdaS2Adj =
-2*g2^4 + 3*lambdaH2Adj^2 - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2 -
(224*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 -
(4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1])/3 -
(8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3 - (4*Bar[lambdaT3GeneralQ000013I1]*Tr[y2 .
Trans[Bar[y2Conj]]])/3 + lambdaT3GeneralQ000013I1*((-44*Bar[lambdaT3GeneralQ000013I1])/3 - (4*Tr[y2Conj .
Trans[Bar[y2]]])/3) + lambdaS2Adj*(4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (2*Tr[y2 .
Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3 + (76*Tr[y2 . Trans[Bar[y2]] . y2Conj . Trans[Bar[y2Conj]]])/45 + (76*Tr[y2 .
Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2]]])/45 - (2*Tr[y2Conj . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2Conj]]])/3

beta_y2Conj =
2*Tr[y2Conj . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y2Conj][gen[$i], heavy[$j]])/2 -
(3*gY^2*Matrix[y2Conj][gen[$i], heavy[$j]])/2 + Tr[y2 . Trans[Bar[y2]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + Tr[y2Conj
. Trans[Bar[y2Conj]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + (6*Matrix[y1, Trans[Bar[y1]], y2Conj][gen[$i], heavy[$j]] +
29*Matrix[y2, Trans[Bar[y2]], y2Conj][gen[$i], heavy[$j]] + 29*Matrix[y2Conj, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] +
 18*Matrix[y2Conj, Trans[Bar[y2Conj]], y2Conj][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]], y2Conj][gen[$i],
heavy[$j]])/12

beta_lambdaT3GeneralQ000004I1 =
-24*g2^2*lambdaT3GeneralQ000004I1 + 12*lambdaS2*lambdaT3GeneralQ000004I1 - 8*lambdaS2Adj*lambdaT3GeneralQ000004I1 +
11*lambdaT3GeneralQ000013I1^2 + lambdaT3GeneralQ001102I1^2 + 2*lambdaT3GeneralQ110002I1^2 +
2*lambdaT3GeneralQ000013I1*Tr[y2 . Trans[Bar[y2Conj]]] + lambdaT3GeneralQ000004I1*(4*Tr[y2 . Trans[Bar[y2]]] +
4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (23*Tr[y2 . Trans[Bar[y2Conj]] . y2 . Trans[Bar[y2Conj]]])/15

beta_lambdaT3GeneralQ000013I1 =
-24*g2^2*lambdaT3GeneralQ000013I1 + 2*lambda12*lambdaT3GeneralQ001102I1 + 4*lambdaH2*lambdaT3GeneralQ110002I1 +
lambdaS2*(22*lambdaT3GeneralQ000013I1 + 2*Tr[y2 . Trans[Bar[y2Conj]]]) +
lambdaT3GeneralQ000004I1*(88*Bar[lambdaT3GeneralQ000013I1] + 8*Tr[y2Conj . Trans[Bar[y2]]]) +
lambdaT3GeneralQ000013I1*(4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (46*Tr[y2 . Trans[Bar[y2]] .
 y2 . Trans[Bar[y2Conj]]])/15 - (46*Tr[y2 . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2Conj]]])/15

beta_lambdaT3GeneralQ001102I1 =
-12*g2^2*lambdaT3GeneralQ001102I1 - 6*gY^2*lambdaT3GeneralQ001102I1 + 4*lambdaS1*lambdaT3GeneralQ001102I1 +
2*lambdaS2*lambdaT3GeneralQ001102I1 - 4*lambdaS2Adj*lambdaT3GeneralQ001102I1 + 4*lambdaT3*lambdaT3GeneralQ020101I1 +
4*lambdaH1*lambdaT3GeneralQ110002I1 + 40*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ001102I1] +
lambda12*(10*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ001102I1 + 2*Tr[y2 . Trans[Bar[y2Conj]]]) +
lambdaT3GeneralQ001102I1*(4*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2Conj]]]

beta_lambdaT3GeneralQ020101I1 =
(-21*g2^2*lambdaT3GeneralQ020101I1)/2 - (9*gY^2*lambdaT3GeneralQ020101I1)/2 + 2*lambda12*lambdaT3GeneralQ020101I1 +
2*lambdaH*lambdaT3GeneralQ020101I1 + 4*lambdaH1*lambdaT3GeneralQ020101I1 + 4*lambdaH2*lambdaT3GeneralQ020101I1 -
4*lambdaH2Adj*lambdaT3GeneralQ020101I1 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3] +
8*lambdaT3GeneralQ110002I1*Bar[lambdaT3] + 2*Bar[lambdaT3]*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ020101I1*(2*Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + Tr[y2Conj . Trans[Bar[y2Conj]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110002I1 =
2*lambdaH1*lambdaT3GeneralQ001102I1 + 4*lambdaT3*lambdaT3GeneralQ020101I1 - (33*g2^2*lambdaT3GeneralQ110002I1)/2 -
(3*gY^2*lambdaT3GeneralQ110002I1)/2 + 6*lambdaH*lambdaT3GeneralQ110002I1 + 2*lambdaS2*lambdaT3GeneralQ110002I1 -
4*lambdaS2Adj*lambdaT3GeneralQ110002I1 + 40*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ110002I1] +
lambdaH2*(10*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ110002I1 + 2*Tr[y2 . Trans[Bar[y2Conj]]]) +
lambdaT3GeneralQ110002I1*(2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]]

beta_muT3GeneralS2Sq =
2*lambdaT3GeneralQ001102I1*mS1Sq - 12*g2^2*muT3GeneralS2Sq + 2*lambdaS2*muT3GeneralS2Sq - 4*lambdaS2Adj*muT3GeneralS2Sq
 + 40*lambdaT3GeneralQ000004I1*Bar[muT3GeneralS2Sq] + mS2Sq*(10*lambdaT3GeneralQ000013I1 + 2*Tr[y2 .
Trans[Bar[y2Conj]]]) + muT3GeneralS2Sq*(2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 4*Tr[MF .
Trans[Bar[y2Conj]] . y2 . Trans[Bar[MF]]]
```

MODEL A_-4

Extra quartics: 0; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
-(Bar[Field[NewScalar1, Scalar, {}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {}, {}]) + Bar[Field[H,
Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1,
SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG,
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H,
Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {},
{}] + (Bar[Field[NewScalar1, Scalar, {}, {}]]^2*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {}, {}]^2)/2 +
Bar[Field[NewScalar1, Scalar, {}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] - Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] +
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
```

```
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar1, Scalar, {},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$2, T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar,
 {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2,
T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv2, {}, 0]*Field[NewScalar2,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 - Coupling[MF, {},
0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*Coupling[y2, {Index[d$$1,
Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG,
{Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]*Field[NewScalar1, Scalar, {}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG],
{Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {}, {}]]*CG[Bar[T3Y1CG],
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(163*gY^3)/6

beta_g2 =
-1/2*g2^3

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (2*Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 6*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + 2*Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + (3*Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]])/4 + Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]]/2

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/2

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (27*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + 2*Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]])/4

beta_mS1Sq =
-24*gY^2*mS1Sq + 6*lambda12*mS2Sq + mS1Sq*(4*lambdaS1 + 4*Tr[y1 . Trans[Bar[y1]]]) - 8*Tr[MF . Trans[Bar[MF]] .
```

```
Trans[y1] . Bar[y1]]

beta_mS2Sq =
2*lambda12*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + mS2Sq*(8*lambdaS2 + 4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]]) -
4*Tr[MF . Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 2*lambdaH1^2 + 6*lambdaH2^2 +
 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] +
12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye .
Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
192*gY^4 + 6*lambda12^2 + 4*lambdaH1^2 - 48*gY^2*lambdaS1 + 10*lambdaS1^2 + 8*lambdaS1*Tr[y1 . Trans[Bar[y1]]] -
8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]]

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH2^2 + gY^2*(8*g2^2 - 12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 +
16*lambdaS2Adj^2 + lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 . Trans[Bar[y2]]]) - (8*Tr[y2 . Trans[Bar[y2]] . y2 .
Trans[Bar[y2]]])/3

beta_lambdaH1 =
12*gY^4 - (9*g2^2*lambdaH1)/2 - (51*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambda12*lambdaH2 +
4*lambdaH1*lambdaS1 + 12*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + 2*lambda12*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2
 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 .
Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 .
Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
48*gY^4 - 12*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 4*lambda12*lambdaS1 +
8*lambda12*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + lambda12*(4*lambdaS2Adj + 4*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 .
Trans[Bar[y2]]]) - 4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*(4*lambdaH2Adj + 2*Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH2Adj =
gY^2*(-4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj + 8*lambdaH2*lambdaH2Adj +
2*lambdaH2Adj*lambdaS2 + (8*lambdaT3*Bar[lambdaT3])/3 + lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (4*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]])/3

beta_lambdaS2Adj =
-2*g2^4 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj +
20*lambdaS2Adj^2 + 4*lambdaS2Adj*Tr[y2 . Trans[Bar[y2]]] - (2*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3
```

MODEL B_+1

Extra quartics: 1; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
Bar[Coupling[lambdaT3ExtraQ022000I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ022000I1], {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
 Lorentz]}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]],
Bar[Index[d$$2, T3BSMd2]], Index[d$$1, T3BSMd2]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
 {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ022000I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaT3ExtraQ022000I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd2],
Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS1Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 -
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] + Bar[Coupling[lambdaT3, {},
```

```
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv1,
 {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]],
Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1,
T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS2Inv1, {},
0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 -
Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion,
{Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3Y2CG, {Index[d$$1,
SU2L[fund]], Index[d$$1, T3BSMd2]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion,
{Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y1,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*NCM[Bar[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]],
Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]]]] + I*NCM[Bar[Field[NewFermion, Fermion,
{Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1,
NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG],
 {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion,
{Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]],
Index[d$$1, T3BSMd2]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(25*gY^3)/2

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/4 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/2

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/4 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/2

beta_MF =
-6*gY^2*Matrix[MF][heavy[$i], heavy[$j]] + Matrix[MF, Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1],
Bar[y1], MF][heavy[$i], heavy[$j]]

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 4*lambda12*mS2Sq + mS1Sq*(6*lambdaS1 + 2*Tr[y1 . Trans[Bar[y1]]]) - 4*Tr[MF .
Trans[Bar[MF]] . Trans[y1] . Bar[y1]]
```

```
beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (27*gY^2*mS2Sq)/2 + mS2Sq*(6*lambdaS2 + 2*Tr[y2 . Trans[Bar[y2]]]) - 4*Tr[MF .
Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 +
lambdaH1Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] +
8*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] +
4*lambdaS1*Tr[y1 . Trans[Bar[y1]]] - 4*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]]

beta_lambdaS2 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((27*g2^2)/2 -
27*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 4*lambdaS2*Tr[y2 . Trans[Bar[y2]]] - 4*Tr[y2 . Trans[Bar[y2]] . y2 .
Trans[Bar[y2]]]

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)/4 +
 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] +
24*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] + lambdaH1*(2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
(9*g2^4)/4 + (27*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 15*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2
 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] + lambdaH2*(2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd
 . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 +
4*lambdaH1*lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3] + lambda12*(2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - 2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + Tr[y1 . Trans[Bar[y1]]] + Tr[y2 .
Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 +
64*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] + lambdaH1Adj*(2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (8*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(18*g2^2 - 15*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] + lambdaH2Adj*(2*Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + 8*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3 + lambda12Adj*(2*Tr[y1 . Trans[Bar[y1]]]
 + 2*Tr[y2 . Trans[Bar[y2]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaT3GeneralQ022000I1 =
-9*g2^2*lambdaT3GeneralQ022000I1 - 3*gY^2*lambdaT3GeneralQ022000I1 + 2*lambdaH*lambdaT3GeneralQ022000I1 +
8*lambdaH1*lambdaT3GeneralQ022000I1 + 4*lambdaH1Adj*lambdaT3GeneralQ022000I1 + 2*lambdaS1*lambdaT3GeneralQ022000I1 +
lambdaT3GeneralQ022000I1*(2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]])
```

MODEL B_-1

Extra quartics: 7; extra bilinears: 1; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3ExtraCGM000101I1, {Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Coupling[lambdaT3ExtraM000101I1, {}, 0] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]},
{}]]*CG[T3ExtraCGQ000202I1, {Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2], Index[d$$3, T3BSMd2], Index[d$$4,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ000202I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3ExtraCGQ020200I1, {Index[d$$1, SU2L[fund]],
Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Coupling[lambdaT3ExtraQ020200I1, {}, 0] +
```

```
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3ExtraCGQ110101I1, {Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Coupling[lambdaT3ExtraQ110101I1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3ExtraCGQ110101I2,
{Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ110101I2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] +
Bar[Coupling[lambdaT3ExtraQ020020I1, {}, 0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ020020I1], {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
 Lorentz]}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]],
Bar[Index[d$$2, T3BSMd2]], Index[d$$1, T3BSMd2]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
 {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Coupling[lambdaT3ExtraQ020200I1, {}, 0]]*CG[Bar[T3ExtraCGQ020200I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2,
 SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]},
{}]]*CG[T3ExtraCGQ001201I1, {Bar[Index[d$$4, T3BSMd2]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2], Index[d$$3,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ001201I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd2]}, {}] +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 - Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Coupling[MS2, {},
0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1,
SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1,
SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv2, {},
 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Coupling[lambdaT3ExtraM000101I1, {}, 0]]*CG[Bar[T3ExtraCGM000101I1], {Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2,
T3BSMd2]]}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]
 + Bar[Coupling[lambdaT3ExtraQ110101I1, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[Bar[T3ExtraCGQ110101I1], {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Coupling[lambdaT3ExtraQ110101I2, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[Bar[T3ExtraCGQ110101I2], {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2,
T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ020020I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaT3ExtraQ020020I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]},
{}]]*CG[T3ExtraCGQ000112I1, {Index[d$$1, T3BSMd2], Bar[Index[d$$4, T3BSMd2]], Index[d$$2, T3BSMd2], Index[d$$3,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ000112I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]],
Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Coupling[lambdaT3ExtraQ001201I1, {},
0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ001201I1], {Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]], Bar[Index[d$$3, T3BSMd2]], Bar[Index[d$$4, T3BSMd2]]}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$4, T3BSMd2]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar,
 {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2,
T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 + Bar[Coupling[lambdaT3ExtraQ000112I1, {},
0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ000112I1], {Bar[Index[d$$2, T3BSMd2]],
Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Bar[Index[d$$4, T3BSMd2]]}]*Field[NewScalar1, Scalar, {Index[d$$2,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]},
{}] + Bar[Coupling[lambdaT3ExtraQ000202I1, {}, 0]]*CG[Bar[T3ExtraCGQ000202I1], {Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]], Bar[Index[d$$3, T3BSMd2]], Bar[Index[d$$4, T3BSMd2]]}]*Field[NewScalar1, Scalar,
```

```
{Index[d$$1, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2]}]*Coupling[y2, {Index[d$$1,
Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd2]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]] - (Coupling[MF, {},
0]*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaCC], Field[NewFermion,
Fermion, {Index[d$$1, NFlavor]}, {}]])/2 + (I/2)*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaCC, GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {Index[d$$1,
Lorentz]}]] + Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1,
T3BSMd2]}]*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaCC, Proj[-1]],
Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y2, {Index[d$$1, Flavor],
Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]]}]*Field[NewScalar2,
 Scalar, {Index[d$$1, T3BSMd2]}, {}]*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaCC, Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(43*gY^3)/6

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/4 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/4 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + Tr[y1 . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (3*Matrix[y1,
 Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 2*Matrix[y1, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/2

beta_y2 =
Tr[y2 . Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/4 -
(3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/4 + Tr[y2 . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (Matrix[y1,
Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 2*Matrix[y2, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/2

beta_MF =
Matrix[Trans[MF], Trans[Bar[y1]], y1][heavy[$i], heavy[$j]] + Matrix[Trans[MF], Trans[Bar[y2]], y2][heavy[$i],
heavy[$j]] + Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]] + Matrix[Trans[y2], Bar[y2], MF][heavy[$i],
heavy[$j]]

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 4*lambda12*mS2Sq + 6*muT3GeneralS1S2*Bar[lambdaS1S1bar2S2bar] +
6*lambdaS1S1bar2S2bar*Bar[muT3GeneralS1S2] + mS1Sq*(6*lambdaS1 + 2*Tr[y1 . Trans[Bar[y1]]]) - 8*Tr[MF . Bar[MF] .
Trans[y1] . Bar[y1]] - 8*Tr[MF . Trans[Bar[y1]] . y1 . Bar[MF]]

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + 6*muT3GeneralS1S2*Bar[lambdaS1barS2S2bar2] +
6*lambdaS1barS2S2bar2*Bar[muT3GeneralS1S2] + mS2Sq*(6*lambdaS2 + 2*Tr[y2 . Trans[Bar[y2]]]) - 8*Tr[MF . Bar[MF] .
Trans[y2] . Bar[y2]] - 8*Tr[MF . Trans[Bar[y2]] . y2 . Bar[MF]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 +
lambdaH1Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
4*lambdaHHdagS1barS2barCross*Bar[lambdaHHdagS1barS2barCross] + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] +
4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ110101I1*Bar[lambdaT3GeneralQ110101I1] + lambdaH*(12*Tr[yd .
Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd .
Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]
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```
beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
8*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 24*lambdaS1S1bar2S2bar*Bar[lambdaS1S1bar2S2bar] + 4*lambdaS1*Tr[y1 .
Trans[Bar[y1]]] - 4*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]]

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] +
8*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 24*lambdaS1barS2S2bar2*Bar[lambdaS1barS2S2bar2] + 4*lambdaS2*Tr[y2 .
Trans[Bar[y2]]] - 4*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]]

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)/4 +
 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 24*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
6*lambdaT3*Bar[lambdaT3] + 3*lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1]) +
lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1S1bar2S2bar] +
2*Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] -
3*Bar[lambdaS1S1bar2S2bar] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH1*(2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
 (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 24*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 6*lambdaT3*Bar[lambdaT3] +
3*lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1]) +
lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1barS2S2bar2] +
2*Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] -
3*Bar[lambdaS1barS2S2bar2] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2*(2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambda12 - 3*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*lambdaH2
+ 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 24*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] +
9*lambdaS1barS2S2bar2*(Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) +
9*lambdaS1S1bar2S2bar*(Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) + 3*lambdaT3*Bar[lambdaT3] +
lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) +
lambdaT3GeneralQ110101I1*(Bar[lambdaHHdagS1barS2barCross] + 2*Bar[lambdaT3GeneralQ110101I1]) + lambda12*(2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - 2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]] - 4*Tr[y1 .
Trans[Bar[y2]] . y2 . Trans[Bar[y1]]]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 3*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 2*Bar[lambdaHHdagS2S2]*Tr[y1 . Trans[Bar[y2]]] - (2*Bar[lambdaHHdagS1barS1bar]*Tr[y2 .
Trans[Bar[y1]]])/3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + Tr[y1 . Trans[Bar[y1]]] + Tr[y2 .
Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - 64*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
4*lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1S1bar2S2bar]) +
(16*lambdaT3*Bar[lambdaT3])/3 + 4*lambdaT3GeneralQ110101I1*(Bar[lambdaS1S1bar2S2bar] - 2*Bar[lambdaT3GeneralQ110101I1])
 + 4*lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) + lambdaH1Adj*(2*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (8*Tr[y1 .
Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 64*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] -
4*lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1barS2S2bar2]) + 16*lambdaT3*Bar[lambdaT3]
 - 4*lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) +
lambdaT3GeneralQ110101I1*(-4*Bar[lambdaS1barS2S2bar2] + 8*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2Adj*(2*Tr[y2 .
Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + 8*Tr[y2 .
Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12Adj =
gY^2*(-6*g2^2 - 3*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + 4*lambdaT3GeneralQ110101I1*Bar[lambdaHHdagS1barS2barCross] -
64*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] - 4*lambdaS1barS2S2bar2*(5*Bar[lambdaS1barS2S2bar2] +
Bar[lambdaS1S1bar2S2bar]) - 4*lambdaS1S1bar2S2bar*(Bar[lambdaS1barS2S2bar2] + 5*Bar[lambdaS1S1bar2S2bar]) +
(4*lambdaT3*Bar[lambdaT3])/3 + 4*lambdaHHdagS1barS2barCross*Bar[lambdaT3GeneralQ110101I1] + lambda12Adj*(2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaHHdagS2S2 =
-9*g2^2*lambdaHHdagS2S2 - 3*gY^2*lambdaHHdagS2S2 + 2*lambdaH*lambdaHHdagS2S2 + 8*lambdaH2*lambdaHHdagS2S2 +
4*lambdaH2Adj*lambdaHHdagS2S2 + 2*lambdaHHdagS2S2*lambdaS2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaS1bar2S2bar2] +
Bar[lambdaT3]*Tr[y1 . Trans[Bar[y2]]] + lambdaHHdagS2S2*(2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaHHdagS1barS1bar =
```

```
-9*g2^2*lambdaHHdagS1barS1bar - 3*gY^2*lambdaHHdagS1barS1bar + 2*lambdaH*lambdaHHdagS1barS1bar +
8*lambdaH1*lambdaHHdagS1barS1bar - 4*lambdaH1Adj*lambdaHHdagS1barS1bar + 2*lambdaHHdagS1barS1bar*lambdaS1 +
4*lambdaHHdagS2S2*lambdaS1bar2S2bar2 - (Bar[lambdaT3]*Tr[y2 . Trans[Bar[y1]]])/3 + lambdaHHdagS1barS1bar*(2*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1bar2S2bar2 =
lambdaHHdagS1barS2barCross^2 - 9*g2^2*lambdaS1bar2S2bar2 - 3*gY^2*lambdaS1bar2S2bar2 + 8*lambda12*lambdaS1bar2S2bar2 -
4*lambda12Adj*lambdaS1bar2S2bar2 + 2*lambdaS1*lambdaS1bar2S2bar2 + 5*lambdaS1barS2S2bar2^2 +
2*lambdaS1barS2S2bar2*lambdaS1S1bar2S2bar + 5*lambdaS1S1bar2S2bar^2 + 2*lambdaS1bar2S2bar2*lambdaS2 +
lambdaT3GeneralQ110101I1^2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS2S2] + lambdaS1bar2S2bar2*(2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) + (2*Tr[y2 . Trans[Bar[y1]] . y2 . Trans[Bar[y1]]])/3

beta_lambdaS1barS2S2bar2 =
-9*g2^2*lambdaS1barS2S2bar2 - 3*gY^2*lambdaS1barS2S2bar2 + lambda12Adj*((-5*lambdaS1barS2S2bar2)/2 -
lambdaS1S1bar2S2bar/2) + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar) + 12*lambdaS1barS2S2bar2*lambdaS2 +
lambdaH2*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH2Adj*(-1/2*lambdaHHdagS1barS2barCross -
lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(5*Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) +
lambdaS1barS2S2bar2*(Tr[y1 . Trans[Bar[y1]]] + 3*Tr[y2 . Trans[Bar[y2]]])

beta_lambdaS1S1bar2S2bar =
lambda12Adj*(-1/2*lambdaS1barS2S2bar2 - (5*lambdaS1S1bar2S2bar)/2) - 9*g2^2*lambdaS1S1bar2S2bar -
3*gY^2*lambdaS1S1bar2S2bar + 12*lambdaS1*lambdaS1S1bar2S2bar + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar)
 + lambdaH1*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH1Adj*(lambdaHHdagS1barS2barCross/2 +
lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(Bar[lambdaS1barS2S2bar2] + 5*Bar[lambdaS1S1bar2S2bar]) +
lambdaS1S1bar2S2bar*(3*Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]])

beta_lambdaHHdagS1barS2barCross =
-9*g2^2*lambdaHHdagS1barS2barCross - 3*gY^2*lambdaHHdagS1barS2barCross + 2*lambda12*lambdaHHdagS1barS2barCross +
lambdaH2*(4*lambdaHHdagS1barS2barCross + 6*lambdaS1barS2S2bar2) + lambdaH1*(4*lambdaHHdagS1barS2barCross +
6*lambdaS1S1bar2S2bar) + lambdaH*(4*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) +
lambdaH2Adj*(-2*lambdaHHdagS1barS2barCross - lambdaS1barS2S2bar2/2 - lambdaT3GeneralQ110101I1) +
lambda12Adj*(-1/2*lambdaHHdagS1barS2barCross + lambdaT3GeneralQ110101I1) + lambdaH1Adj*(2*lambdaHHdagS1barS2barCross +
lambdaS1S1bar2S2bar/2 + lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(8*Bar[lambdaHHdagS1barS2barCross] -
4*Bar[lambdaT3GeneralQ110101I1]) + lambdaHHdagS1barS2barCross*(Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110101I1 =
lambdaH1Adj*(-lambdaHHdagS1barS2barCross + lambdaS1S1bar2S2bar/2 - 2*lambdaT3GeneralQ110101I1) +
lambda12Adj*(lambdaHHdagS1barS2barCross - lambdaT3GeneralQ110101I1/2) - 9*g2^2*lambdaT3GeneralQ110101I1 -
3*gY^2*lambdaT3GeneralQ110101I1 + 2*lambda12*lambdaT3GeneralQ110101I1 + lambdaH2Adj*(lambdaHHdagS1barS2barCross -
lambdaS1barS2S2bar2/2 + 2*lambdaT3GeneralQ110101I1) + lambdaH*(-2*lambdaHHdagS1barS2barCross +
4*lambdaT3GeneralQ110101I1) + lambdaH2*(-6*lambdaS1barS2S2bar2 + 4*lambdaT3GeneralQ110101I1) +
lambdaH1*(-6*lambdaS1S1bar2S2bar + 4*lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(-4*Bar[lambdaHHdagS1barS2barCross]
 + 8*Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1S2 =
6*lambdaS1S1bar2S2bar*mS1Sq + 6*lambdaS1barS2S2bar2*mS2Sq - (9*g2^2*muT3GeneralS1S2)/2 - (3*gY^2*muT3GeneralS1S2)/2 +
2*lambda12*muT3GeneralS1S2 - (3*lambda12Adj*muT3GeneralS1S2)/2 + 12*lambdaS1bar2S2bar2*Bar[muT3GeneralS1S2] +
muT3GeneralS1S2*(Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]])
```

MODEL B_-3

Extra quartics: 1; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3ExtraCGQ020002I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ020002I1, {}, 0] - Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
Lorentz]}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
 {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]],
Bar[Index[d$$2, T3BSMd2]], Index[d$$1, T3BSMd2]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
 {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 - Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Coupling[MS2, {},
0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1,
SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
```

```
Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1,
SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv2, {},
 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Coupling[lambdaT3ExtraQ020002I1, {}, 0]]*CG[Bar[T3ExtraCGQ020002I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2,
 SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd2],
Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd2]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2,
 Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]],
Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 - Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion,
 Fermion, {Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2,
 Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2]}]*Coupling[y2,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1,
Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1,
NFlavor]}, {}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1,
Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1,
 Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1,
T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1,
NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] +
Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
 {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1,
NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(25*gY^3)/2

beta_g2 =
(-17*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 3*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/2

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/4 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/2

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/4 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/2

beta_MF =
-6*gY^2*Matrix[MF][heavy[$i], heavy[$j]] + Matrix[MF, Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1],
Bar[y1], MF][heavy[$i], heavy[$j]]

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (27*gY^2*mS1Sq)/2 + 4*lambda12*mS2Sq + mS1Sq*(6*lambdaS1 + 2*Tr[y1 . Trans[Bar[y1]]]) - 4*Tr[MF .
Trans[Bar[MF]] . Trans[y1] . Bar[y1]]
```

```
beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + mS2Sq*(6*lambdaS2 + 2*Tr[y2 . Trans[Bar[y2]]]) - 4*Tr[MF .
Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 +
lambdaH1Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] +
8*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((27*g2^2)/2 -
27*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 4*lambdaS1*Tr[y1 . Trans[Bar[y1]]] - 4*Tr[y1 . Trans[Bar[y1]] . y1 .
Trans[Bar[y1]]]

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] +
4*lambdaS2*Tr[y2 . Trans[Bar[y2]]] - 4*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]]

beta_lambdaH1 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambdaH1 - 15*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)/4
 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] + lambdaH1*(2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd
 . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
 (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] +
24*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] + lambdaH2*(2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 +
4*lambdaH1*lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3] + lambda12*(2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - 2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + Tr[y1 . Trans[Bar[y1]]] + Tr[y2 .
Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-18*g2^2 - 15*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(2*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (8*Tr[y1 .
Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(-6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] -
64*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] + lambdaH2Adj*(2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + 8*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3 + lambda12Adj*(2*Tr[y1 . Trans[Bar[y1]]]
 + 2*Tr[y2 . Trans[Bar[y2]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaT3GeneralQ020002I1 =
-9*g2^2*lambdaT3GeneralQ020002I1 - 3*gY^2*lambdaT3GeneralQ020002I1 + 2*lambdaH*lambdaT3GeneralQ020002I1 +
8*lambdaH2*lambdaT3GeneralQ020002I1 - 4*lambdaH2Adj*lambdaT3GeneralQ020002I1 + 2*lambdaS2*lambdaT3GeneralQ020002I1 +
lambdaT3GeneralQ020002I1*(2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]])
```

MODEL C_+1

Extra quartics: 1; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
Bar[Coupling[lambdaT3ExtraQ022000I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ022000I1], {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
 Lorentz]}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
```

```
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]],
Bar[Index[d$$2, T3BSMd2]], Index[d$$1, T3BSMd2]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
 {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ022000I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaT3ExtraQ022000I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd2],
Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS1Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 -
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] + Bar[Coupling[lambdaT3, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv1,
 {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]],
Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1,
T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS2Inv1, {},
0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 -
Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$1,
T3BSMd2]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3],
Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Bar[Index[d$$1,
T3BSMd2]]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC,
Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]]] +
I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]},
{Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG],
{Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd3], Bar[Index[d$$1, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$1,
 T3BSMd2]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]],
DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$1, T3BSMd2]}]*NCM[Transp[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(41*gY^3)/2

beta_g2 =
(31*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
```

```
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 2*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-33*g2^2*Matrix[y1][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/4 + (3*Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]])/2 + (5*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] +
3*Matrix[y2, Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-33*g2^2*Matrix[y2][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/4 + (3*Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]])/2 + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] +
5*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
-12*g2^2*Matrix[MF][heavy[$i], heavy[$j]] - 6*gY^2*Matrix[MF][heavy[$i], heavy[$j]] + (Matrix[MF, Trans[Bar[y2]],
y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 4*lambda12*mS2Sq + mS1Sq*(6*lambdaS1 + 3*Tr[y1 . Trans[Bar[y1]]]) - 6*Tr[MF .
Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (27*gY^2*mS2Sq)/2 + mS2Sq*(6*lambdaS2 + 3*Tr[y2 . Trans[Bar[y2]]]) - 6*Tr[MF .
Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 +
lambdaH1Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] +
8*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] +
6*lambdaS1*Tr[y1 . Trans[Bar[y1]]] - 5*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]]

beta_lambdaS2 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((27*g2^2)/2 -
27*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 6*lambdaS2*Tr[y2 . Trans[Bar[y2]]] - 5*Tr[y2 . Trans[Bar[y2]] . y2 .
Trans[Bar[y2]]]

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)/4 +
 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] +
24*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] + lambdaH1*(3*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 3*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
(9*g2^4)/4 + (27*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 15*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2
 + (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] + lambdaH2*(3*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd
 . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 3*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 +
4*lambdaH1*lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3] + lambda12*(3*Tr[y1 .
Trans[Bar[y1]]] + 3*Tr[y2 . Trans[Bar[y2]]]) - (9*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/2

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + (3*Tr[y1 . Trans[Bar[y1]]])/2 +
(3*Tr[y2 . Trans[Bar[y2]]])/2 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 +
64*lambdaT3GeneralQ022000I1*Bar[lambdaT3GeneralQ022000I1] + lambdaH1Adj*(3*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(18*g2^2 - 15*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] + lambdaH2Adj*(3*Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3 + lambda12Adj*(3*Tr[y1 . Trans[Bar[y1]]]
 + 3*Tr[y2 . Trans[Bar[y2]]]) - (2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3
```

```
beta_lambdaT3GeneralQ022000I1 =
-9*g2^2*lambdaT3GeneralQ022000I1 - 3*gY^2*lambdaT3GeneralQ022000I1 + 2*lambdaH*lambdaT3GeneralQ022000I1 +
8*lambdaH1*lambdaT3GeneralQ022000I1 + 4*lambdaH1Adj*lambdaT3GeneralQ022000I1 + 2*lambdaS1*lambdaT3GeneralQ022000I1 +
lambdaT3GeneralQ022000I1*(3*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]])
```

MODEL C_-1

Extra quartics: 7; extra bilinears: 1; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3ExtraCGM000101I1, {Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Coupling[lambdaT3ExtraM000101I1, {}, 0] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]},
{}]]*CG[T3ExtraCGQ000202I1, {Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2], Index[d$$3, T3BSMd2], Index[d$$4,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ000202I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3ExtraCGQ020200I1, {Index[d$$1, SU2L[fund]],
Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Coupling[lambdaT3ExtraQ020200I1, {}, 0] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3ExtraCGQ110101I1, {Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Coupling[lambdaT3ExtraQ110101I1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3ExtraCGQ110101I2,
{Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ110101I2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] +
Bar[Coupling[lambdaT3ExtraQ020020I1, {}, 0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ020020I1], {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2]}]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1,
 Lorentz]}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]],
Bar[Index[d$$2, T3BSMd2]], Index[d$$1, T3BSMd2]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
 {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Coupling[lambdaT3ExtraQ020200I1, {}, 0]]*CG[Bar[T3ExtraCGQ020200I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2,
 SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]},
{}]]*CG[T3ExtraCGQ001201I1, {Bar[Index[d$$4, T3BSMd2]], Index[d$$1, T3BSMd2], Index[d$$2, T3BSMd2], Index[d$$3,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ001201I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd2]}, {}] +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 - Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Coupling[MS2, {},
0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1,
SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1,
SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv2, {},
 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] +
Bar[Coupling[lambdaT3ExtraM000101I1, {}, 0]]*CG[Bar[T3ExtraCGM000101I1], {Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2,
T3BSMd2]]}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]
 + Bar[Coupling[lambdaT3ExtraQ110101I1, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[Bar[T3ExtraCGQ110101I1], {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Coupling[lambdaT3ExtraQ110101I2, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[Bar[T3ExtraCGQ110101I2], {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2,
T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ020020I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaT3ExtraQ020020I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]},
{}]]*CG[T3ExtraCGQ000112I1, {Index[d$$1, T3BSMd2], Bar[Index[d$$4, T3BSMd2]], Index[d$$2, T3BSMd2], Index[d$$3,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ000112I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] +
```

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Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4,
T3BSMd2]]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]],
Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Coupling[lambdaT3ExtraQ001201I1, {},
0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ001201I1], {Index[d$$1, T3BSMd2],
Bar[Index[d$$2, T3BSMd2]], Bar[Index[d$$3, T3BSMd2]], Bar[Index[d$$4, T3BSMd2]]}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$4, T3BSMd2]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar,
 {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2,
T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 + Bar[Coupling[lambdaT3ExtraQ000112I1, {},
0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3ExtraCGQ000112I1], {Bar[Index[d$$2, T3BSMd2]],
Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Bar[Index[d$$4, T3BSMd2]]}]*Field[NewScalar1, Scalar, {Index[d$$2,
T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]},
{}] + Bar[Coupling[lambdaT3ExtraQ000202I1, {}, 0]]*CG[Bar[T3ExtraCGQ000202I1], {Bar[Index[d$$1, T3BSMd2]],
Bar[Index[d$$2, T3BSMd2]], Bar[Index[d$$3, T3BSMd2]], Bar[Index[d$$4, T3BSMd2]]}]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd2]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$1,
T3BSMd2]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3],
Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Bar[Index[d$$1,
T3BSMd2]]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd2]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]] - (Coupling[MF, {},
0]*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaCC],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]])/2 + (I/2)*NCM[Transp[Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaCC, GammaM[Index[d$$1, Lorentz]]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y1,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
{}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$1,
T3BSMd2]}]*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaCC, Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] +
Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]],
Index[d$$1, T3BSMd3], Bar[Index[d$$1, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]},
{}]*NCM[Transp[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaCC,
Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(43*gY^3)/6

beta_g2 =
(7*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 2*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-33*g2^2*Matrix[y1][gen[$i], heavy[$j]])/4 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/4 + (3*Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]])/2 + (3*Tr[y1 . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]])/2 +
(5*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 2*Matrix[y1, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] +
3*Matrix[y2, Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(3*Tr[y2 . Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]])/2 - (33*g2^2*Matrix[y2][gen[$i], heavy[$j]])/4 -
(3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/4 + (3*Tr[y2 . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]])/2 +
(3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 2*Matrix[y2, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] +
5*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
```

```
-6*g2^2*(Matrix[MF][heavy[$i], heavy[$j]] + Matrix[Trans[MF]][heavy[$i], heavy[$j]]) + (Matrix[Trans[MF],
Trans[Bar[y1]], y1][heavy[$i], heavy[$j]] + Matrix[Trans[MF], Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] +
Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]] + Matrix[Trans[y2], Bar[y2], MF][heavy[$i], heavy[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (3*gY^2*mS1Sq)/2 + 4*lambda12*mS2Sq + 6*muT3GeneralS1S2*Bar[lambdaS1S1bar2S2bar] +
6*lambdaS1S1bar2S2bar*Bar[muT3GeneralS1S2] + mS1Sq*(6*lambdaS1 + 3*Tr[y1 . Trans[Bar[y1]]]) - 12*Tr[MF . Bar[MF] .
Trans[y1] . Bar[y1]] - 12*Tr[MF . Trans[Bar[y1]] . y1 . Bar[MF]]

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + 6*muT3GeneralS1S2*Bar[lambdaS1barS2S2bar2] +
6*lambdaS1barS2S2bar2*Bar[muT3GeneralS1S2] + mS2Sq*(6*lambdaS2 + 3*Tr[y2 . Trans[Bar[y2]]]) - 12*Tr[MF . Bar[MF] .
Trans[y2] . Bar[y2]] - 12*Tr[MF . Trans[Bar[y2]] . y2 . Bar[MF]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 +
lambdaH1Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
4*lambdaHHdagS1barS2barCross*Bar[lambdaHHdagS1barS2barCross] + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] +
4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ110101I1*Bar[lambdaT3GeneralQ110101I1] + lambdaH*(12*Tr[yd .
Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd .
Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 8*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
8*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 24*lambdaS1S1bar2S2bar*Bar[lambdaS1S1bar2S2bar] + 6*lambdaS1*Tr[y1 .
Trans[Bar[y1]]] - 5*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]]

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] +
8*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] + 24*lambdaS1barS2S2bar2*Bar[lambdaS1barS2S2bar2] + 6*lambdaS2*Tr[y2 .
Trans[Bar[y2]]] - 5*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]]

beta_lambdaH1 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambdaH1 - 3*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)/4 +
 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 24*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
6*lambdaT3*Bar[lambdaT3] + 3*lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1]) +
lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1S1bar2S2bar] +
2*Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] -
3*Bar[lambdaS1S1bar2S2bar] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH1*(3*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 3*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
 (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 24*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] + 6*lambdaT3*Bar[lambdaT3] +
3*lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] - Bar[lambdaT3GeneralQ110101I1]) +
lambdaHHdagS1barS2barCross*(4*Bar[lambdaHHdagS1barS2barCross] + 3*Bar[lambdaS1barS2S2bar2] +
2*Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*(2*Bar[lambdaHHdagS1barS2barCross] -
3*Bar[lambdaS1barS2S2bar2] + 4*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2*(3*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 3*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(9*g2^4)/4 + (3*gY^4)/4 - 9*g2^2*lambda12 - 3*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 + 4*lambdaH1*lambdaH2
+ 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 24*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] +
9*lambdaS1barS2S2bar2*(Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) +
9*lambdaS1S1bar2S2bar*(Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) + 3*lambdaT3*Bar[lambdaT3] +
lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) +
lambdaT3GeneralQ110101I1*(Bar[lambdaHHdagS1barS2barCross] + 2*Bar[lambdaT3GeneralQ110101I1]) + lambda12*(3*Tr[y1 .
Trans[Bar[y1]]] + 3*Tr[y2 . Trans[Bar[y2]]]) - (9*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/2 - 3*Tr[y1 .
Trans[Bar[y2]] . y2 . Trans[Bar[y1]]]

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 3*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 3*Bar[lambdaHHdagS2S2]*Tr[y1 . Trans[Bar[y2]]] - Bar[lambdaHHdagS1barS1bar]*Tr[y2 .
Trans[Bar[y1]]] + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + (3*Tr[y1 . Trans[Bar[y1]]])/2 +
(3*Tr[y2 . Trans[Bar[y2]]])/2 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-6*g2^2 - 3*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - 64*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS1barS1bar] +
4*lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1S1bar2S2bar]) +
(16*lambdaT3*Bar[lambdaT3])/3 + 4*lambdaT3GeneralQ110101I1*(Bar[lambdaS1S1bar2S2bar] - 2*Bar[lambdaT3GeneralQ110101I1])
 + 4*lambdaS1S1bar2S2bar*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) + lambdaH1Adj*(3*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 .
Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj +
```

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8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 64*lambdaHHdagS2S2*Bar[lambdaHHdagS2S2] -
4*lambdaHHdagS1barS2barCross*(2*Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaS1barS2S2bar2]) + 16*lambdaT3*Bar[lambdaT3]
 - 4*lambdaS1barS2S2bar2*(Bar[lambdaHHdagS1barS2barCross] + Bar[lambdaT3GeneralQ110101I1]) +
lambdaT3GeneralQ110101I1*(-4*Bar[lambdaS1barS2S2bar2] + 8*Bar[lambdaT3GeneralQ110101I1]) + lambdaH2Adj*(3*Tr[y2 .
Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y2 .
Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12Adj =
gY^2*(-6*g2^2 - 3*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + 4*lambdaT3GeneralQ110101I1*Bar[lambdaHHdagS1barS2barCross] -
64*lambdaS1bar2S2bar2*Bar[lambdaS1bar2S2bar2] - 4*lambdaS1barS2S2bar2*(5*Bar[lambdaS1barS2S2bar2] +
Bar[lambdaS1S1bar2S2bar]) - 4*lambdaS1S1bar2S2bar*(Bar[lambdaS1barS2S2bar2] + 5*Bar[lambdaS1S1bar2S2bar]) +
(4*lambdaT3*Bar[lambdaT3])/3 + 4*lambdaHHdagS1barS2barCross*Bar[lambdaT3GeneralQ110101I1] + lambda12Adj*(3*Tr[y1 .
Trans[Bar[y1]]] + 3*Tr[y2 . Trans[Bar[y2]]]) - (2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3 - (8*Tr[y1 .
Trans[Bar[y2]] . y2 . Trans[Bar[y1]]])/3

beta_lambdaHHdagS2S2 =
-9*g2^2*lambdaHHdagS2S2 - 3*gY^2*lambdaHHdagS2S2 + 2*lambdaH*lambdaHHdagS2S2 + 8*lambdaH2*lambdaHHdagS2S2 +
4*lambdaH2Adj*lambdaHHdagS2S2 + 2*lambdaHHdagS2S2*lambdaS2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaS1bar2S2bar2] +
(3*Bar[lambdaT3]*Tr[y1 . Trans[Bar[y2]]])/2 + lambdaHHdagS2S2*(3*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaHHdagS1barS1bar =
-9*g2^2*lambdaHHdagS1barS1bar - 3*gY^2*lambdaHHdagS1barS1bar + 2*lambdaH*lambdaHHdagS1barS1bar +
8*lambdaH1*lambdaHHdagS1barS1bar - 4*lambdaH1Adj*lambdaHHdagS1barS1bar + 2*lambdaHHdagS1barS1bar*lambdaS1 +
4*lambdaHHdagS2S2*lambdaS1bar2S2bar2 - (Bar[lambdaT3]*Tr[y2 . Trans[Bar[y1]]])/2 + lambdaHHdagS1barS1bar*(3*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaS1bar2S2bar2 =
lambdaHHdagS1barS2barCross^2 - 9*g2^2*lambdaS1bar2S2bar2 - 3*gY^2*lambdaS1bar2S2bar2 + 8*lambda12*lambdaS1bar2S2bar2 -
4*lambda12Adj*lambdaS1bar2S2bar2 + 2*lambdaS1*lambdaS1bar2S2bar2 + 5*lambdaS1barS2S2bar2^2 +
2*lambdaS1barS2S2bar2*lambdaS1S1bar2S2bar + 5*lambdaS1S1bar2S2bar^2 + 2*lambdaS1bar2S2bar2*lambdaS2 +
lambdaT3GeneralQ110101I1^2 + 4*lambdaHHdagS1barS1bar*Bar[lambdaHHdagS2S2] + lambdaS1bar2S2bar2*(3*Tr[y1 .
Trans[Bar[y1]]] + 3*Tr[y2 . Trans[Bar[y2]]]) + (5*Tr[y2 . Trans[Bar[y1]] . y2 . Trans[Bar[y1]]])/6

beta_lambdaS1barS2S2bar2 =
-9*g2^2*lambdaS1barS2S2bar2 - 3*gY^2*lambdaS1barS2S2bar2 + lambda12Adj*((-5*lambdaS1barS2S2bar2)/2 -
lambdaS1S1bar2S2bar/2) + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar) + 12*lambdaS1barS2S2bar2*lambdaS2 +
lambdaH2*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH2Adj*(-1/2*lambdaHHdagS1barS2barCross -
lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(5*Bar[lambdaS1barS2S2bar2] + Bar[lambdaS1S1bar2S2bar]) +
lambdaS1barS2S2bar2*((3*Tr[y1 . Trans[Bar[y1]]])/2 + (9*Tr[y2 . Trans[Bar[y2]]])/2)

beta_lambdaS1S1bar2S2bar =
lambda12Adj*(-1/2*lambdaS1barS2S2bar2 - (5*lambdaS1S1bar2S2bar)/2) - 9*g2^2*lambdaS1S1bar2S2bar -
3*gY^2*lambdaS1S1bar2S2bar + 12*lambdaS1*lambdaS1S1bar2S2bar + lambda12*(6*lambdaS1barS2S2bar2 + 6*lambdaS1S1bar2S2bar)
 + lambdaH1*(2*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) + lambdaH1Adj*(lambdaHHdagS1barS2barCross/2 +
lambdaT3GeneralQ110101I1/2) + 4*lambdaS1bar2S2bar2*(Bar[lambdaS1barS2S2bar2] + 5*Bar[lambdaS1S1bar2S2bar]) +
lambdaS1S1bar2S2bar*((9*Tr[y1 . Trans[Bar[y1]]])/2 + (3*Tr[y2 . Trans[Bar[y2]]])/2)

beta_lambdaHHdagS1barS2barCross =
-9*g2^2*lambdaHHdagS1barS2barCross - 3*gY^2*lambdaHHdagS1barS2barCross + 2*lambda12*lambdaHHdagS1barS2barCross +
lambdaH2*(4*lambdaHHdagS1barS2barCross + 6*lambdaS1barS2S2bar2) + lambdaH1*(4*lambdaHHdagS1barS2barCross +
6*lambdaS1S1bar2S2bar) + lambdaH*(4*lambdaHHdagS1barS2barCross - 2*lambdaT3GeneralQ110101I1) +
lambdaH2Adj*(-2*lambdaHHdagS1barS2barCross - lambdaS1barS2S2bar2/2 - lambdaT3GeneralQ110101I1) +
lambda12Adj*(-1/2*lambdaHHdagS1barS2barCross + lambdaT3GeneralQ110101I1) + lambdaH1Adj*(2*lambdaHHdagS1barS2barCross +
lambdaS1S1bar2S2bar/2 + lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(8*Bar[lambdaHHdagS1barS2barCross] -
4*Bar[lambdaT3GeneralQ110101I1]) + lambdaHHdagS1barS2barCross*((3*Tr[y1 . Trans[Bar[y1]]])/2 + (3*Tr[y2 .
Trans[Bar[y2]]])/2 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110101I1 =
lambdaH1Adj*(-lambdaHHdagS1barS2barCross + lambdaS1S1bar2S2bar/2 - 2*lambdaT3GeneralQ110101I1) +
lambda12Adj*(lambdaHHdagS1barS2barCross - lambdaT3GeneralQ110101I1/2) - 9*g2^2*lambdaT3GeneralQ110101I1 -
3*gY^2*lambdaT3GeneralQ110101I1 + 2*lambda12*lambdaT3GeneralQ110101I1 + lambdaH2Adj*(lambdaHHdagS1barS2barCross -
lambdaS1barS2S2bar2/2 + 2*lambdaT3GeneralQ110101I1) + lambdaH*(-2*lambdaHHdagS1barS2barCross +
4*lambdaT3GeneralQ110101I1) + lambdaH2*(-6*lambdaS1barS2S2bar2 + 4*lambdaT3GeneralQ110101I1) +
lambdaH1*(-6*lambdaS1S1bar2S2bar + 4*lambdaT3GeneralQ110101I1) + lambdaS1bar2S2bar2*(-4*Bar[lambdaHHdagS1barS2barCross]
 + 8*Bar[lambdaT3GeneralQ110101I1]) + lambdaT3GeneralQ110101I1*((3*Tr[y1 . Trans[Bar[y1]]])/2 + (3*Tr[y2 .
Trans[Bar[y2]]])/2 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_muT3GeneralS1S2 =
6*lambdaS1S1bar2S2bar*mS1Sq + 6*lambdaS1barS2S2bar2*mS2Sq - (9*g2^2*muT3GeneralS1S2)/2 - (3*gY^2*muT3GeneralS1S2)/2 +
2*lambda12*muT3GeneralS1S2 - (3*lambda12Adj*muT3GeneralS1S2)/2 + 12*lambdaS1bar2S2bar2*Bar[muT3GeneralS1S2] +
muT3GeneralS1S2*((3*Tr[y1 . Trans[Bar[y1]]])/2 + (3*Tr[y2 . Trans[Bar[y2]]])/2)
```

MODEL C_-3

Extra quartics: 1; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]},
{}]]*CG[T3ExtraCGQ020002I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$2,
T3BSMd2]}]*Coupling[lambdaT3ExtraQ020002I1, {}, 0] - Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]},
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{}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Bar[Index[d$$2, T3BSMd2]], Index[d$$1, T3BSMd2]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 - Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {Index[d$$1, Lorentz]}] + Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd2], Bar[Index[d$$2, T3BSMd2]]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}] + Bar[Coupling[lambdaT3ExtraQ020002I1, {}, 0]]*CG[Bar[T3ExtraCGQ020002I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Bar[Index[d$$2, T3BSMd2]]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd2]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd2], Bar[Index[d$$3, T3BSMd2]], Index[d$$2, T3BSMd2], Bar[Index[d$$4, T3BSMd2]]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd2]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd2]}, {}])/2 - Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$1, T3BSMd2]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd3], Bar[Index[d$$1, T3BSMd2]]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd2]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd2]}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$1, T3BSMd2]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd3], Index[d$$1, NFlavor]}, {}]]

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(41*gY^3)/2

beta_g2 =
(31*g2^3)/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2
```

```
beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 2*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-33*g2^2*Matrix[y1][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/4 + (3*Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]])/2 + (5*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] +
3*Matrix[y2, Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-33*g2^2*Matrix[y2][gen[$i], heavy[$j]])/4 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/4 + (3*Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]])/2 + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] +
5*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
-12*g2^2*Matrix[MF][heavy[$i], heavy[$j]] - 6*gY^2*Matrix[MF][heavy[$i], heavy[$j]] + (Matrix[MF, Trans[Bar[y2]],
y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]])/2

beta_mS1Sq =
(-9*g2^2*mS1Sq)/2 - (27*gY^2*mS1Sq)/2 + 4*lambda12*mS2Sq + mS1Sq*(6*lambdaS1 + 3*Tr[y1 . Trans[Bar[y1]]]) - 6*Tr[MF .
Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
4*lambda12*mS1Sq - (9*g2^2*mS2Sq)/2 - (3*gY^2*mS2Sq)/2 + mS2Sq*(6*lambdaS2 + 3*Tr[y2 . Trans[Bar[y2]]]) - 6*Tr[MF .
Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 4*lambdaH1^2 +
lambdaH1Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + 4*lambdaT3*Bar[lambdaT3] +
8*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
(9*g2^4)/4 + (243*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH1^2 + lambdaH1Adj^2/4 + gY^2*((27*g2^2)/2 -
27*lambdaS1) - 9*g2^2*lambdaS1 + 12*lambdaS1^2 + 6*lambdaS1*Tr[y1 . Trans[Bar[y1]]] - 5*Tr[y1 . Trans[Bar[y1]] . y1 .
Trans[Bar[y1]]]

beta_lambdaS2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12^2 + lambda12Adj^2/4 + 4*lambdaH2^2 + lambdaH2Adj^2/4 + gY^2*((3*g2^2)/2 -
3*lambdaS2) - 9*g2^2*lambdaS2 + 12*lambdaS2^2 + 8*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] +
6*lambdaS2*Tr[y2 . Trans[Bar[y2]]] - 5*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]]

beta_lambdaH1 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambdaH1 - 15*gY^2*lambdaH1 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + (3*lambdaH1Adj^2)/4
 + 4*lambda12*lambdaH2 + 6*lambdaH1*lambdaS1 + 6*lambdaT3*Bar[lambdaT3] + lambdaH1*(3*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd
 . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 3*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
(9*g2^4)/4 + (3*gY^4)/4 + 4*lambda12*lambdaH1 - 9*g2^2*lambdaH2 - 3*gY^2*lambdaH2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
 (3*lambdaH2Adj^2)/4 + 6*lambdaH2*lambdaS2 + 6*lambdaT3*Bar[lambdaT3] +
24*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] + lambdaH2*(3*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 3*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(9*g2^4)/4 + (27*gY^4)/4 - 9*g2^2*lambda12 - 15*gY^2*lambda12 + 4*lambda12^2 + (3*lambda12Adj^2)/4 +
4*lambdaH1*lambdaH2 + 6*lambda12*lambdaS1 + 6*lambda12*lambdaS2 + 3*lambdaT3*Bar[lambdaT3] + lambda12*(3*Tr[y1 .
Trans[Bar[y1]]] + 3*Tr[y2 . Trans[Bar[y2]]]) - (9*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/2

beta_lambdaT3 =
-9*g2^2*lambdaT3 - 9*gY^2*lambdaT3 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*(lambda12Adj/6 + (2*lambdaH1Adj)/3 + 2*lambdaH2Adj + (3*Tr[y1 . Trans[Bar[y1]]])/2 +
(3*Tr[y2 . Trans[Bar[y2]]])/2 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-18*g2^2 - 15*lambdaH1Adj) - 9*g2^2*lambdaH1Adj + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 + (16*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(3*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 .
Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
lambda12Adj*lambdaH1Adj + gY^2*(-6*g2^2 - 3*lambdaH2Adj) - 9*g2^2*lambdaH2Adj + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 + 16*lambdaT3*Bar[lambdaT3] -
64*lambdaT3GeneralQ020002I1*Bar[lambdaT3GeneralQ020002I1] + lambdaH2Adj*(3*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]
```

```
beta_lambda12Adj =
gY^2*(18*g2^2 - 15*lambda12Adj) - 9*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 2*lambda12Adj*lambdaS2 + (4*lambdaT3*Bar[lambdaT3])/3 + lambda12Adj*(3*Tr[y1 . Trans[Bar[y1]]]
 + 3*Tr[y2 . Trans[Bar[y2]]]) - (2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaT3GeneralQ020002I1 =
-9*g2^2*lambdaT3GeneralQ020002I1 - 3*gY^2*lambdaT3GeneralQ020002I1 + 2*lambdaH*lambdaT3GeneralQ020002I1 +
8*lambdaH2*lambdaT3GeneralQ020002I1 - 4*lambdaH2Adj*lambdaT3GeneralQ020002I1 + 2*lambdaS2*lambdaT3GeneralQ020002I1 +
lambdaT3GeneralQ020002I1*(3*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]])
```

MODEL D_+0

Extra quartics: 5; extra bilinears: 1; extra Yukawa: y1Conj

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGM000200I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraM000200I1, {}, 0] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000400I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ000400I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGQ110200I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ110200I1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] +
Bar[Coupling[lambdaT3ExtraQ021010I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[Bar[T3ExtraCGQ021010I1], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2,
SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1,
 Scalar, {Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3MixCG, {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar,
{Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 +
Bar[Coupling[lambdaT3ExtraM000200I1, {}, 0]]*CG[T3ExtraCGM000200I1, {Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]
+ Bar[Coupling[lambdaT3ExtraQ110200I1, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ110200I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]},
{}]]*CG[T3ExtraCGQ001300I1, {Index[d$$4, T3BSMd3], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ001300I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv2, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 +
Bar[Coupling[lambdaT3ExtraQ001300I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3ExtraCGQ001300I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ000400I1, {},
0]]*CG[T3ExtraCGQ000400I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1,
SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*Bar[Field[NewScalar2,
Scalar, {}, {}]]*CG[T3ExtraCGQ000211I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraQ000211I1,
{}, 0]*Field[NewScalar2, Scalar, {}, {}] - Bar[Field[NewScalar2, Scalar, {}, {}]]*Coupling[MS2, {},
0]^2*Field[NewScalar2, Scalar, {}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2,
Scalar, {}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2,
SU2L[fund]]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ021010I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ021010I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
```

```
{}]*Field[NewScalar2, Scalar, {}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {}, {}] + Bar[Coupling[lambdaT3ExtraQ000211I1, {}, 0]]*Bar[Field[NewScalar2, Scalar, {},
{}]]*CG[T3ExtraCGQ000211I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {}, {}] +
(Bar[Field[NewScalar2, Scalar, {}, {}]]^2*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {}, {}]^2)/2 +
Bar[Field[NewScalar2, Scalar, {}, {Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {}, {Index[d$$1, Lorentz]}] -
Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {},
{}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1,
NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3Y1ConjCG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$1,
T3BSMd3]}]*Coupling[y1Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2],
Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG],
{Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2]}]*Field[NewScalar2, Scalar, {}, {}]*NCM[Bar[Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]],
Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1,
Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]] + Bar[Coupling[y1Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y1ConjCG],
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1,
 T3BSMd3]}, {}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(55*gY^3)/6

beta_g2 =
-1/2*g2^3

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + 3*Matrix[y1Conj,
Trans[Bar[y1Conj]], ye][gen[$i], gen[$j]] + 2*Matrix[y2, Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 6*Matrix[ye,
Trans[Bar[ye]], ye][gen[$i], gen[$j]])/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + Tr[y1Conj . Trans[Bar[y1Conj]]]*Matrix[y1][gen[$i], heavy[$j]] +
2*Tr[y1 . Trans[Bar[y1Conj]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + (18*Matrix[y1, Trans[Bar[y1]], y1][gen[$i],
heavy[$j]] + 29*Matrix[y1, Trans[Bar[y1Conj]], y1Conj][gen[$i], heavy[$j]] + 29*Matrix[y1Conj, Trans[Bar[y1Conj]],
y1][gen[$i], heavy[$j]] + 6*Matrix[y2, Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]],
y1][gen[$i], heavy[$j]])/12

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + 2*Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] +
3*Matrix[y1Conj, Trans[Bar[y1Conj]], y2][gen[$i], heavy[$j]] + 4*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] +
2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (3*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (2*Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + 3*(Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]] +
Matrix[Trans[y1Conj], Bar[y1Conj], MF][heavy[$i], heavy[$j]]))/4
```

```
beta_mS1Sq =
-12*g2^2*mS1Sq + 2*lambda12*mS2Sq + 20*lambdaT3GeneralQ001300I1*Bar[muT3GeneralS1Sq] +
muT3GeneralS1Sq*(20*Bar[lambdaT3GeneralQ001300I1] + 4*Tr[y1 . Trans[Bar[y1Conj]]]) + 4*Bar[muT3GeneralS1Sq]*Tr[y1Conj .
 Trans[Bar[y1]]] + mS1Sq*(8*lambdaS1 + 4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]]) -
 4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]] - 4*Tr[MF . Trans[Bar[MF]] . Trans[y1Conj] . Bar[y1Conj]]

beta_mS2Sq =
6*lambda12*mS1Sq - 6*gY^2*mS2Sq + 12*muT3GeneralS1Sq*Bar[lambdaT3GeneralQ000211I1] +
12*lambdaT3GeneralQ000211I1*Bar[muT3GeneralS1Sq] + mS2Sq*(4*lambdaS2 + 4*Tr[y2 . Trans[Bar[y2]]]) - 8*Tr[MF .
Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 2*lambdaH2^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] +
24*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 2*lambda12^2 + 4*lambdaH1^2 - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 16*lambdaS1Adj^2 +
(8*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1])/3 +
(832*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1])/3 +
(16*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3 +
lambdaT3GeneralQ001300I1*((352*Bar[lambdaT3GeneralQ001300I1])/3 + (32*Tr[y1 . Trans[Bar[y1Conj]]])/3) +
(32*Bar[lambdaT3GeneralQ001300I1]*Tr[y1Conj . Trans[Bar[y1]]])/3 + lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 . Trans[Bar[y1]]]
+ 4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3 - (248*Tr[y1 .
Trans[Bar[y1]] . y1Conj . Trans[Bar[y1Conj]]])/45 - (248*Tr[y1 . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1]]])/45 -
(8*Tr[y1Conj . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1Conj]]])/3

beta_lambdaS2 =
12*gY^4 + 6*lambda12^2 + 4*lambdaH2^2 - 12*gY^2*lambdaS2 + 10*lambdaS2^2 +
24*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1] + 8*lambdaS2*Tr[y2 . Trans[Bar[y2]]] - 8*Tr[y2 .
Trans[Bar[y2]] . y2 . Trans[Bar[y2]]]

beta_lambdaH1 =
6*g2^4 - (33*g2^2*lambdaH1)/2 - (3*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 +
2*lambda12*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 20*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ110200I1] +
lambdaT3GeneralQ110200I1*(4*(5*Bar[lambdaT3GeneralQ001300I1] + 4*Bar[lambdaT3GeneralQ110200I1]) + 4*Tr[y1 .
Trans[Bar[y1Conj]]]) + 4*Bar[lambdaT3GeneralQ110200I1]*Tr[y1Conj . Trans[Bar[y1]]] + lambdaH1*(4*lambdaS1Adj + 2*Tr[y1
. Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu
 . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]] - 2*Tr[y1Conj . Trans[Bar[y1Conj]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
3*gY^4 + 6*lambda12*lambdaH1 - (9*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
4*lambdaH2*lambdaS2 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ000211I1] +
12*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 12*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ110200I1] +
 lambdaH2*(4*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]]) - 4*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 8*lambda12*lambdaS1 + 4*lambda12*lambdaS2 +
4*lambdaT3*Bar[lambdaT3] + 20*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ000211I1] +
4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + lambdaT3GeneralQ000211I1*(4*(4*Bar[lambdaT3GeneralQ000211I1]
 + 5*Bar[lambdaT3GeneralQ001300I1]) + 4*Tr[y1 . Trans[Bar[y1Conj]]]) + 4*Bar[lambdaT3GeneralQ000211I1]*Tr[y1Conj .
Trans[Bar[y1]]] + lambda12*(4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 4*Tr[y2 .
Trans[Bar[y2]]]) - 4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]] - 4*Tr[y1Conj . Trans[Bar[y1Conj]] . y2 .
Trans[Bar[y2]]]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 4*Bar[lambdaT3GeneralQ021010I1]*(Bar[lambdaT3GeneralQ000211I1] + 2*Bar[lambdaT3GeneralQ110200I1])
 + 2*Bar[lambdaT3GeneralQ021010I1]*Tr[y1 . Trans[Bar[y1Conj]]] + lambdaT3*(-4*lambdaH1Adj + Tr[y1 . Trans[Bar[y1]]] +
Tr[y1Conj . Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
(-33*g2^2*lambdaH1Adj)/2 - (3*gY^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
2*lambdaH1Adj*lambdaS1 - (8*lambdaT3*Bar[lambdaT3])/3 + (8*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1])/3 +
lambdaH1Adj*(16*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]]
 + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3 +
(4*Tr[y1Conj . Trans[Bar[y1Conj]] . ye . Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 3*lambdaH1Adj^2 - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2 -
(4*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I1])/3 -
(224*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1])/3 -
(8*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3 +
lambdaT3GeneralQ001300I1*((-44*Bar[lambdaT3GeneralQ001300I1])/3 - (4*Tr[y1 . Trans[Bar[y1Conj]]])/3) -
(4*Bar[lambdaT3GeneralQ001300I1]*Tr[y1Conj . Trans[Bar[y1]]])/3 + lambdaS1Adj*(4*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y1Conj
```

```
. Trans[Bar[y1Conj]]]) - (2*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3 + (76*Tr[y1 . Trans[Bar[y1]] . y1Conj .
Trans[Bar[y1Conj]]])/45 + (76*Tr[y1 . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1]]])/45 - (2*Tr[y1Conj .
Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1Conj]]])/3

beta_y1Conj =
2*Tr[y1Conj . Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y1Conj][gen[$i], heavy[$j]])/2 -
(3*gY^2*Matrix[y1Conj][gen[$i], heavy[$j]])/2 + Tr[y1 . Trans[Bar[y1]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + Tr[y1Conj
. Trans[Bar[y1Conj]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + (29*Matrix[y1, Trans[Bar[y1]], y1Conj][gen[$i], heavy[$j]] +
 29*Matrix[y1Conj, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 6*(3*Matrix[y1Conj, Trans[Bar[y1Conj]], y1Conj][gen[$i],
heavy[$j]] + Matrix[y2, Trans[Bar[y2]], y1Conj][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y1Conj][gen[$i],
heavy[$j]]))/12

beta_lambdaT3GeneralQ000211I1 =
-12*g2^2*lambdaT3GeneralQ000211I1 - 6*gY^2*lambdaT3GeneralQ000211I1 + 2*lambdaS1*lambdaT3GeneralQ000211I1 -
4*lambdaS1Adj*lambdaT3GeneralQ000211I1 + 4*lambdaS2*lambdaT3GeneralQ000211I1 + 4*lambdaH2*lambdaT3GeneralQ110200I1 +
40*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000211I1] + 4*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] +
lambda12*(8*lambdaT3GeneralQ000211I1 + 10*lambdaT3GeneralQ001300I1 + 2*Tr[y1Conj . Trans[Bar[y1]]]) +
lambdaT3GeneralQ000211I1*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 4*Tr[y2 . Trans[Bar[y2]]]) -
4*Tr[y1Conj . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]]

beta_lambdaT3GeneralQ000400I1 =
lambdaT3GeneralQ000211I1^2 - 24*g2^2*lambdaT3GeneralQ000400I1 + 12*lambdaS1*lambdaT3GeneralQ000400I1 -
8*lambdaS1Adj*lambdaT3GeneralQ000400I1 + 11*lambdaT3GeneralQ001300I1^2 + 2*lambdaT3GeneralQ110200I1^2 +
2*lambdaT3GeneralQ001300I1*Tr[y1Conj . Trans[Bar[y1]]] + lambdaT3GeneralQ000400I1*(4*Tr[y1 . Trans[Bar[y1]]] +
4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - (23*Tr[y1Conj . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1]]])/15

beta_lambdaT3GeneralQ001300I1 =
2*lambda12*lambdaT3GeneralQ000211I1 - 24*g2^2*lambdaT3GeneralQ001300I1 + 4*lambdaH1*lambdaT3GeneralQ110200I1 +
lambdaT3GeneralQ000400I1*(88*Bar[lambdaT3GeneralQ001300I1] + 8*Tr[y1 . Trans[Bar[y1Conj]]]) +
lambdaS1*(22*lambdaT3GeneralQ001300I1 + 2*Tr[y1Conj . Trans[Bar[y1]]]) + lambdaT3GeneralQ001300I1*(4*Tr[y1 .
Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - (46*Tr[y1 . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1]]])/15 -
(46*Tr[y1Conj . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1Conj]]])/15

beta_lambdaT3GeneralQ021010I1 =
(-21*g2^2*lambdaT3GeneralQ021010I1)/2 - (9*gY^2*lambdaT3GeneralQ021010I1)/2 + 2*lambda12*lambdaT3GeneralQ021010I1 +
2*lambdaH*lambdaT3GeneralQ021010I1 + 4*lambdaH1*lambdaT3GeneralQ021010I1 + 4*lambdaH1Adj*lambdaT3GeneralQ021010I1 +
4*lambdaH2*lambdaT3GeneralQ021010I1 + 4*Bar[lambdaT3]*(Bar[lambdaT3GeneralQ000211I1] + 2*Bar[lambdaT3GeneralQ110200I1])
 + 2*Bar[lambdaT3]*Tr[y1 . Trans[Bar[y1Conj]]] + lambdaT3GeneralQ021010I1*(Tr[y1 . Trans[Bar[y1]]] + Tr[y1Conj .
Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110200I1 =
2*lambdaH2*lambdaT3GeneralQ000211I1 - (33*g2^2*lambdaT3GeneralQ110200I1)/2 - (3*gY^2*lambdaT3GeneralQ110200I1)/2 +
6*lambdaH*lambdaT3GeneralQ110200I1 + 2*lambdaS1*lambdaT3GeneralQ110200I1 - 4*lambdaS1Adj*lambdaT3GeneralQ110200I1 +
4*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] + 40*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ110200I1] +
lambdaH1*(10*lambdaT3GeneralQ001300I1 + 8*lambdaT3GeneralQ110200I1 + 2*Tr[y1Conj . Trans[Bar[y1]]]) +
lambdaT3GeneralQ110200I1*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1Conj . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_muT3GeneralS1Sq =
2*lambdaT3GeneralQ000211I1*mS2Sq - 12*g2^2*muT3GeneralS1Sq + 2*lambdaS1*muT3GeneralS1Sq - 4*lambdaS1Adj*muT3GeneralS1Sq
 + 40*lambdaT3GeneralQ000400I1*Bar[muT3GeneralS1Sq] + mS1Sq*(10*lambdaT3GeneralQ001300I1 + 2*Tr[y1Conj .
Trans[Bar[y1]]]) + muT3GeneralS1Sq*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]]) - 4*Tr[MF .
Trans[Bar[MF]] . Trans[y1Conj] . Bar[y1]]
```

MODEL D_+2

Extra quartics: 0; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
-(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]) + Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]
 + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, 
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
```

```
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv2, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + Bar[Coupling[lambdaT3,
{}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {}, {}] - Bar[Field[NewScalar2, Scalar, {},
{}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar, {}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2,
SU2L[fund]]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {},
{}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambda12Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {}, {}] + (Bar[Field[NewScalar2,
Scalar, {}, {}]]^2*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {}, {}]^2)/2 + Bar[Field[NewScalar2, Scalar,
 {}, {Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {}, {Index[d$$1, Lorentz]}] - Coupling[MF, {},
0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3Y2CG, {Index[d$$1,
SU2L[fund]], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2],
Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG],
{Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2]}]*Field[NewScalar2, Scalar, {}, {}]*NCM[Bar[Field[NewFermion,
Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]],
Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1,
Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(163*gY^3)/6

beta_g2 =
-1/2*g2^3

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + 2*Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 6*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]])/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/2

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + 2*Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]])/4 + Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]]/2

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (27*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (2*Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + 3*Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]])/4

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + 2*lambda12*mS2Sq + mS1Sq*(8*lambdaS1 + 4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]]) -
4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
6*lambda12*mS1Sq - 24*gY^2*mS2Sq + mS2Sq*(4*lambdaS2 + 4*Tr[y2 . Trans[Bar[y2]]]) - 8*Tr[MF . Trans[Bar[y2]] . y2 .
Trans[Bar[MF]]]
```

```
beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 2*lambdaH2^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH1^2 + gY^2*(8*g2^2 - 12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 +
16*lambdaS1Adj^2 + lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 . Trans[Bar[y1]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 .
Trans[Bar[y1]]])/3

beta_lambdaS2 =
192*gY^4 + 6*lambda12^2 + 4*lambdaH2^2 - 48*gY^2*lambdaS2 + 10*lambdaS2^2 + 8*lambdaS2*Tr[y2 . Trans[Bar[y2]]] -
8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]]

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 +
2*lambda12*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*lambdaS1Adj + 2*Tr[y1 .
Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1 .
Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_lambdaH2 =
12*gY^4 + 6*lambda12*lambdaH1 - (9*g2^2*lambdaH2)/2 - (51*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
4*lambdaH2*lambdaS2 + 12*lambdaT3*Bar[lambdaT3] + lambdaH2*(4*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] +
2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
48*gY^4 - 12*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 8*lambda12*lambdaS1 +
4*lambda12*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] + lambda12*(4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y2 .
Trans[Bar[y2]]]) - 4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*(-4*lambdaH1Adj + Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
2*lambdaH1Adj*lambdaS1 - (8*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(16*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj +
20*lambdaS1Adj^2 + 4*lambdaS1Adj*Tr[y1 . Trans[Bar[y1]]] - (2*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3
```

MODEL D_-2

Extra quartics: 5; extra bilinears: 1; extra Yukawa: y2Conj

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar2, Scalar, {}, {}]]^2*Coupling[lambdaT3ExtraM000002I1, {}, 0] + Bar[Field[NewScalar2, Scalar, {},
{}]]^4*Coupling[lambdaT3ExtraQ000004I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3ExtraCGQ020101I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]],
Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3ExtraQ020101I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]^2*CG[T3ExtraCGQ110002I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1,
SU2L[fund]]}]*Coupling[lambdaT3ExtraQ110002I1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] -
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3MixCG, {Bar[Index[d$$1, SU2L[fund]]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]
 + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]^2*CG[T3ExtraCGQ001102I1, {Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ001102I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
(Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
 T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1,
Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2,
T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]},
```

```
{}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv2, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + Bar[Coupling[lambdaT3,
{}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {}, {}] + Bar[Field[NewScalar2, Scalar, {},
{}]]^3*Coupling[lambdaT3ExtraQ000013I1, {}, 0]*Field[NewScalar2, Scalar, {}, {}] - Bar[Field[NewScalar2, Scalar, {},
{}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar, {}, {}] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2,
SU2L[fund]]]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{}, {}] + Bar[Coupling[lambdaT3ExtraQ020101I1, {}, 0]]*CG[Bar[T3ExtraCGQ020101I1], {Bar[Index[d$$1, SU2L[fund]]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3]}]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar,
{Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {}, {}]
+ Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {},
{}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambda12Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {}, {}] +
Bar[Coupling[lambdaT3ExtraM000002I1, {}, 0]]*Field[NewScalar2, Scalar, {}, {}]^2 + (Bar[Field[NewScalar2, Scalar, {},
{}]]^2*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {}, {}]^2)/2 + Bar[Coupling[lambdaT3ExtraQ110002I1, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[T3ExtraCGQ110002I1, {Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, SU2L[fund]]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {}, {}]^2 +
Bar[Coupling[lambdaT3ExtraQ001102I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3ExtraCGQ001102I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2,
T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {}, {}]^2 + Bar[Coupling[lambdaT3ExtraQ000013I1, {}, 0]]*Bar[Field[NewScalar2,
 Scalar, {}, {}]]*Field[NewScalar2, Scalar, {}, {}]^3 + Bar[Coupling[lambdaT3ExtraQ000004I1, {}, 0]]*Field[NewScalar2,
Scalar, {}, {}]^4 + Bar[Field[NewScalar2, Scalar, {}, {Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {},
{Index[d$$1, Lorentz]}] - Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2,
Scalar, {}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y2, {Index[d$$1, Flavor],
Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y2ConjCG,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]]}]*Coupling[y2Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]*Field[NewScalar2, Scalar, {}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG,
{Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1,
NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
 Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1,
T3BSMd2], Index[d$$1, NFlavor]}, {}]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*Bar[Field[NewScalar2, Scalar, {}, {}]]*CG[Bar[T3Y2ConjCG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1,
T3BSMd2]}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y2,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1,
T3BSMd2]}]*Field[NewScalar2, Scalar, {}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] +
Bar[Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
 {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC,
Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(59*gY^3)/6

beta_g2 =
-1/2*g2^3

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + 2*(Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + Matrix[y2Conj, Trans[Bar[y2Conj]], ye][gen[$i], gen[$j]] + 3*Matrix[ye,
Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
```

```
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + Matrix[y2Conj, Trans[Bar[y2Conj]], y1][gen[$i], heavy[$j]] + Matrix[ye,
Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/2

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + 2*Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]*Matrix[y2][gen[$i], heavy[$j]] +
4*Tr[y2 . Trans[Bar[y2Conj]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i],
heavy[$j]] + 2*(2*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 5*Matrix[y2, Trans[Bar[y2Conj]],
y2Conj][gen[$i], heavy[$j]] + 5*Matrix[y2Conj, Trans[Bar[y2Conj]], y2][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]],
 y2][gen[$i], heavy[$j]]))/4

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (3*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (2*Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + 2*Matrix[MF, Trans[Bar[y2Conj]], y2Conj][heavy[$i], heavy[$j]] +
3*Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]])/4

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + 2*lambda12*mS2Sq + 4*muT3GeneralS2Sq*Bar[lambdaT3GeneralQ001102I1] +
4*lambdaT3GeneralQ001102I1*Bar[muT3GeneralS2Sq] + mS1Sq*(8*lambdaS1 + 4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]]) -
4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
6*lambda12*mS1Sq + 12*lambdaT3GeneralQ000013I1*Bar[muT3GeneralS2Sq] + 8*Bar[muT3GeneralS2Sq]*Tr[y2 .
Trans[Bar[y2Conj]]] + muT3GeneralS2Sq*(12*Bar[lambdaT3GeneralQ000013I1] + 8*Tr[y2Conj . Trans[Bar[y2]]]) +
mS2Sq*(4*lambdaS2 + 4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 8*Tr[MF . Trans[Bar[y2]] . y2 .
Trans[Bar[MF]]] - 8*Tr[MF . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 2*lambdaH2^2 + 4*lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] +
8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye .
Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 2*lambda12^2 + 4*lambdaH1^2 + gY^2*(8*g2^2 - 12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 +
16*lambdaS1Adj^2 + 8*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1] + lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 .
Trans[Bar[y1]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_lambdaS2 =
6*lambda12^2 + 4*lambdaH2^2 + 10*lambdaS2^2 + 288*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1] +
12*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I1] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] +
24*Bar[lambdaT3GeneralQ000013I1]*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ000013I1*(108*Bar[lambdaT3GeneralQ000013I1] + 24*Tr[y2Conj . Trans[Bar[y2]]]) + lambdaS2*(8*Tr[y2 .
Trans[Bar[y2]]] + 8*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]] - 16*Tr[y2 .
Trans[Bar[y2]] . y2Conj . Trans[Bar[y2Conj]]] - 16*Tr[y2 . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2]]] - 8*Tr[y2Conj
. Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2Conj]]]

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 +
2*lambda12*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ001102I1] + 4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] +
4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH1*(4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
6*lambda12*lambdaH1 - (9*g2^2*lambdaH2)/2 - (3*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 + 4*lambdaH2^2 +
4*lambdaH2*lambdaS2 + 12*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] +
12*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ110002I1] + 8*Bar[lambdaT3GeneralQ110002I1]*Tr[y2 . Trans[Bar[y2Conj]]]
 + lambdaT3GeneralQ110002I1*(4*(3*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ110002I1]) + 8*Tr[y2Conj .
Trans[Bar[y2]]]) + lambdaH2*(4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]]
+ 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 4*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]] -
4*Tr[y2Conj . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
-12*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 4*lambdaH1*lambdaH2 + 8*lambda12*lambdaS1 + 4*lambda12*lambdaS2 +
4*lambdaT3*Bar[lambdaT3] + 12*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ001102I1] +
4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 8*Bar[lambdaT3GeneralQ001102I1]*Tr[y2 . Trans[Bar[y2Conj]]]
+ lambdaT3GeneralQ001102I1*(4*(3*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ001102I1]) + 8*Tr[y2Conj .
Trans[Bar[y2]]]) + lambda12*(4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj .
Trans[Bar[y2Conj]]]) - 4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]] - 4*Tr[y1 . Trans[Bar[y1]] . y2Conj .
Trans[Bar[y2Conj]]]

beta_lambdaT3 =
(-21*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ020101I1] +
8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ020101I1] + 4*Bar[lambdaT3GeneralQ020101I1]*Tr[y2 . Trans[Bar[y2Conj]]]
+ lambdaT3*(-4*lambdaH1Adj + Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
```

```
beta_lambdaH1Adj =
gY^2*(-4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
2*lambdaH1Adj*lambdaS1 - (8*lambdaT3*Bar[lambdaT3])/3 - (8*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1])/3 +
lambdaH1Adj*(16*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj +
20*lambdaS1Adj^2 + 4*lambdaS1Adj*Tr[y1 . Trans[Bar[y1]]] - (2*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_y2Conj =
4*Tr[y2Conj . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y2Conj][gen[$i], heavy[$j]])/2 -
(3*gY^2*Matrix[y2Conj][gen[$i], heavy[$j]])/2 + 2*Tr[y2 . Trans[Bar[y2]]]*Matrix[y2Conj][gen[$i], heavy[$j]] +
2*Tr[y2Conj . Trans[Bar[y2Conj]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2Conj][gen[$i],
heavy[$j]] + 2*(5*Matrix[y2, Trans[Bar[y2]], y2Conj][gen[$i], heavy[$j]] + 5*Matrix[y2Conj, Trans[Bar[y2]],
y2][gen[$i], heavy[$j]] + 2*Matrix[y2Conj, Trans[Bar[y2Conj]], y2Conj][gen[$i], heavy[$j]] + Matrix[ye, Trans[Bar[ye]],
 y2Conj][gen[$i], heavy[$j]]))/4

beta_lambdaT3GeneralQ000004I1 =
12*lambdaS2*lambdaT3GeneralQ000004I1 + 9*lambdaT3GeneralQ000013I1^2 + 3*lambdaT3GeneralQ001102I1^2 +
2*lambdaT3GeneralQ110002I1^2 + 4*lambdaT3GeneralQ000013I1*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ000004I1*(8*Tr[y2 . Trans[Bar[y2]]] + 8*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 4*Tr[y2 . Trans[Bar[y2Conj]]
 . y2 . Trans[Bar[y2Conj]]]

beta_lambdaT3GeneralQ000013I1 =
6*lambda12*lambdaT3GeneralQ001102I1 + 4*lambdaH2*lambdaT3GeneralQ110002I1 + lambdaS2*(18*lambdaT3GeneralQ000013I1 +
4*Tr[y2 . Trans[Bar[y2Conj]]]) + lambdaT3GeneralQ000004I1*(72*Bar[lambdaT3GeneralQ000013I1] + 16*Tr[y2Conj .
Trans[Bar[y2]]]) + lambdaT3GeneralQ000013I1*(8*Tr[y2 . Trans[Bar[y2]]] + 8*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 8*Tr[y2 .
 Trans[Bar[y2]] . y2 . Trans[Bar[y2Conj]]] - 8*Tr[y2 . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2Conj]]]

beta_lambdaT3GeneralQ001102I1 =
-12*g2^2*lambdaT3GeneralQ001102I1 - 6*gY^2*lambdaT3GeneralQ001102I1 + 8*lambdaS1*lambdaT3GeneralQ001102I1 +
4*lambdaS1Adj*lambdaT3GeneralQ001102I1 + 2*lambdaS2*lambdaT3GeneralQ001102I1 + 4*lambdaT3*lambdaT3GeneralQ020101I1 +
4*lambdaH1*lambdaT3GeneralQ110002I1 + 24*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ001102I1] +
lambda12*(6*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ001102I1 + 4*Tr[y2 . Trans[Bar[y2Conj]]]) +
lambdaT3GeneralQ001102I1*(2*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2Conj]]]

beta_lambdaT3GeneralQ020101I1 =
(-21*g2^2*lambdaT3GeneralQ020101I1)/2 - (9*gY^2*lambdaT3GeneralQ020101I1)/2 + 2*lambda12*lambdaT3GeneralQ020101I1 +
2*lambdaH*lambdaT3GeneralQ020101I1 + 4*lambdaH1*lambdaT3GeneralQ020101I1 - 4*lambdaH1Adj*lambdaT3GeneralQ020101I1 +
4*lambdaH2*lambdaT3GeneralQ020101I1 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3] +
8*lambdaT3GeneralQ110002I1*Bar[lambdaT3] + 4*Bar[lambdaT3]*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ020101I1*(Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110002I1 =
6*lambdaH1*lambdaT3GeneralQ001102I1 + 12*lambdaT3*lambdaT3GeneralQ020101I1 - (9*g2^2*lambdaT3GeneralQ110002I1)/2 -
(3*gY^2*lambdaT3GeneralQ110002I1)/2 + 6*lambdaH*lambdaT3GeneralQ110002I1 + 2*lambdaS2*lambdaT3GeneralQ110002I1 +
24*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH2*(6*lambdaT3GeneralQ000013I1 +
8*lambdaT3GeneralQ110002I1 + 4*Tr[y2 . Trans[Bar[y2Conj]]]) + lambdaT3GeneralQ110002I1*(4*Tr[y2 . Trans[Bar[y2]]] +
4*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
- 4*Tr[y2 . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]]

beta_muT3GeneralS2Sq =
6*lambdaT3GeneralQ001102I1*mS1Sq + 2*lambdaS2*muT3GeneralS2Sq + 24*lambdaT3GeneralQ000004I1*Bar[muT3GeneralS2Sq] +
mS2Sq*(6*lambdaT3GeneralQ000013I1 + 4*Tr[y2 . Trans[Bar[y2Conj]]]) + muT3GeneralS2Sq*(4*Tr[y2 . Trans[Bar[y2]]] +
4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - 8*Tr[MF . Trans[Bar[y2Conj]] . y2 . Trans[Bar[MF]]]
```

MODEL E_+0

Extra quartics: 6; extra bilinears: 1; extra Yukawa: y1Conj

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGM000200I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraM000200I1, {}, 0] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000400I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ000400I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGQ110200I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ110200I1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] +
Bar[Coupling[lambdaT3ExtraQ021010I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[Bar[T3ExtraCGQ021010I1], {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[H, Scalar, {Index[d$$1,
SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1,
 Lorentz]}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3],
Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1,
```

```
Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG,
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraM000200I1, {},
0]]*CG[T3ExtraCGM000200I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ110200I1, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*CG[T3ExtraCGQ110200I1, {Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*CG[T3ExtraCGQ001300I1, {Index[d$$4, T3BSMd3],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3]}]*Coupling[lambdaT3ExtraQ001300I1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv2,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv2, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 +
Bar[Coupling[lambdaT3ExtraQ001300I1, {}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3ExtraCGQ001300I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ000400I1, {},
0]]*CG[T3ExtraCGQ000400I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] -
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + Bar[Coupling[lambdaT3, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
(Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
 T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
 SU2L[fund]]}, {}]]*CG[T3ExtraCGQ021010I1, {Index[d$$1, SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd3],
Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraQ021010I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000211I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3], Index[d$$3,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ000211I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*CG[T3ExtraCGQ000211I2, {Index[d$$1, T3BSMd3],
Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3], Index[d$$3, T3BSMd3]}]*Coupling[lambdaT3ExtraQ000211I2, {},
0]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv2,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv2, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3CrossScalarCGInv3, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambda12Inv3, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ000211I1, {}, 0]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3ExtraCGQ000211I1, {Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$1, T3BSMd3],
Index[d$$4, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$3,
T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ000211I2, {},
0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3ExtraCGQ000211I2, {Index[d$$2, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$1, T3BSMd3], Index[d$$4, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
```

```
{}]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
(Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv2, {}, 0]*Field[NewScalar2,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 - Coupling[MF, {},
0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*Coupling[y2, {Index[d$$1,
Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3Y1ConjCG, {Index[d$$1, SU2L[fund]], Index[d$$1,
T3BSMd2], Index[d$$1, T3BSMd3]}]*Coupling[y1Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l,
Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]],
Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] + CG[T3Y1CG, {Index[d$$1,
SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1,
T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1,
 {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC,
Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Coupling[y1Conj,
{Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y1ConjCG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1,
T3BSMd2]], Index[d$$1, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Transp[Field[l, Fermion,
{Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(59*gY^3)/6

beta_g2 =
g2^3/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y1Conj,
Trans[Bar[y1Conj]], ye][gen[$i], gen[$j]] + Matrix[y2, Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 2*Matrix[ye,
Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + Tr[y1Conj . Trans[Bar[y1Conj]]]*Matrix[y1][gen[$i], heavy[$j]] +
2*Tr[y1 . Trans[Bar[y1Conj]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + (18*Matrix[y1, Trans[Bar[y1]], y1][gen[$i],
heavy[$j]] + 29*Matrix[y1, Trans[Bar[y1Conj]], y1Conj][gen[$i], heavy[$j]] + 29*Matrix[y1Conj, Trans[Bar[y1Conj]],
y1][gen[$i], heavy[$j]] + 9*Matrix[y2, Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]],
y1][gen[$i], heavy[$j]])/12

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] +
3*Matrix[y1Conj, Trans[Bar[y1Conj]], y2][gen[$i], heavy[$j]] + 6*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] +
2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (3*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*(Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]] +
Matrix[Trans[y1Conj], Bar[y1Conj], MF][heavy[$i], heavy[$j]]))/4
```

```
beta_mS1Sq =
-12*g2^2*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq + 20*lambdaT3GeneralQ001300I1*Bar[muT3GeneralS1Sq] +
muT3GeneralS1Sq*(20*Bar[lambdaT3GeneralQ001300I1] + 4*Tr[y1 . Trans[Bar[y1Conj]]]) + 4*Bar[muT3GeneralS1Sq]*Tr[y1Conj .
 Trans[Bar[y1]]] + mS1Sq*(8*lambdaS1 + 4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]]) -
 4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]] - 4*Tr[MF . Trans[Bar[MF]] . Trans[y1Conj] . Bar[y1Conj]]

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq +
4*muT3GeneralS1Sq*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
12*lambdaT3GeneralQ000211I1*Bar[muT3GeneralS1Sq] + 4*lambdaT3GeneralQ000211I2*Bar[muT3GeneralS1Sq] + mS2Sq*(8*lambdaS2
+ 4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]]) - 4*Tr[MF . Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 24*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1] +
lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd .
Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] .
yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 - 24*g2^2*lambdaS1 +
14*lambdaS1^2 + 16*lambdaS1Adj^2 + (8*lambdaT3GeneralQ000211I1*(3*Bar[lambdaT3GeneralQ000211I1] +
Bar[lambdaT3GeneralQ000211I2]))/3 + lambdaT3GeneralQ000211I2*((8*Bar[lambdaT3GeneralQ000211I1])/3 +
4*Bar[lambdaT3GeneralQ000211I2]) + (832*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1])/3 +
(16*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3 +
lambdaT3GeneralQ001300I1*((352*Bar[lambdaT3GeneralQ001300I1])/3 + (32*Tr[y1 . Trans[Bar[y1Conj]]])/3) +
(32*Bar[lambdaT3GeneralQ001300I1]*Tr[y1Conj . Trans[Bar[y1]]])/3 + lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 . Trans[Bar[y1]]]
+ 4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3 - (248*Tr[y1 .
Trans[Bar[y1]] . y1Conj . Trans[Bar[y1Conj]]])/45 - (248*Tr[y1 . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1]]])/45 -
(8*Tr[y1Conj . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1Conj]]])/3

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 + gY^2*(8*g2^2 -
12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 16*lambdaS2Adj^2 +
8*lambdaT3GeneralQ000211I1*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
lambdaT3GeneralQ000211I2*(8*Bar[lambdaT3GeneralQ000211I1] + (20*Bar[lambdaT3GeneralQ000211I2])/3) +
lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 . Trans[Bar[y2]]]) - (8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaH1 =
6*g2^4 - (33*g2^2*lambdaH1)/2 - (3*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 + (6*lambda12
 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 20*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ110200I1] +
lambdaT3GeneralQ110200I1*(4*(5*Bar[lambdaT3GeneralQ001300I1] + 4*Bar[lambdaT3GeneralQ110200I1]) + 4*Tr[y1 .
Trans[Bar[y1Conj]]]) + 4*Bar[lambdaT3GeneralQ110200I1]*Tr[y1Conj . Trans[Bar[y1]]] + lambdaH1*(4*lambdaS1Adj + 2*Tr[y1
. Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu
 . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]] - 2*Tr[y1Conj . Trans[Bar[y1Conj]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 +
6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ110200I1*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
4*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + 12*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ110200I1] +
4*lambdaT3GeneralQ000211I2*Bar[lambdaT3GeneralQ110200I1] + lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]]

beta_lambda12 =
(63*g2^4)/5 - 24*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 +
4*lambdaH1*lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 +
4*lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + 2*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ001300I1*(5*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
4*lambdaT3GeneralQ000211I2*(Bar[lambdaT3GeneralQ000211I2] + Bar[lambdaT3GeneralQ001300I1]) +
2*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] + lambdaT3GeneralQ000211I1*(4*(4*Bar[lambdaT3GeneralQ000211I1]
 + 5*Bar[lambdaT3GeneralQ001300I1]) + 4*Tr[y1 . Trans[Bar[y1Conj]]]) + 4*Bar[lambdaT3GeneralQ000211I1]*Tr[y1Conj .
Trans[Bar[y1]]] + lambda12*(4*lambdaS1Adj + 4*lambdaS2Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj .
Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (22*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/15 -
(22*Tr[y1Conj . Trans[Bar[y1Conj]] . y2 . Trans[Bar[y2]]])/15

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 2*Bar[lambdaT3GeneralQ021010I1]*(2*Bar[lambdaT3GeneralQ000211I1] - Bar[lambdaT3GeneralQ000211I2]
+ 4*Bar[lambdaT3GeneralQ110200I1]) + 2*Bar[lambdaT3GeneralQ021010I1]*Tr[y1 . Trans[Bar[y1Conj]]] +
lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross - 2*lambdaH1Adj - (4*lambdaH2Adj)/3 + Tr[y1 . Trans[Bar[y1]]] +
Tr[y1Conj . Trans[Bar[y1Conj]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] +
6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
(-33*g2^2*lambdaH1Adj)/2 - (3*gY^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
```

```
12*lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 -
(8*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1])/9 + lambdaH1Adj*(16*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]]
+ 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]]) - (4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3 + (4*Tr[y1Conj . Trans[Bar[y1Conj]] . ye .
Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj + gY^2*(4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 -
(8*lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1])/9 + lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]]
+ 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (4*Tr[y2 . Trans[Bar[y2]] . ye .
 Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 - 24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj +
20*lambdaS1Adj^2 - (4*lambdaT3GeneralQ000211I2*Bar[lambdaT3GeneralQ000211I1])/3 -
(4*lambdaT3GeneralQ000211I1*(3*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]))/3 -
(224*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000400I1])/3 -
(8*lambdaT3GeneralQ110200I1*Bar[lambdaT3GeneralQ110200I1])/3 +
lambdaT3GeneralQ001300I1*((-44*Bar[lambdaT3GeneralQ001300I1])/3 - (4*Tr[y1 . Trans[Bar[y1Conj]]])/3) -
(4*Bar[lambdaT3GeneralQ001300I1]*Tr[y1Conj . Trans[Bar[y1]]])/3 + lambdaS1Adj*(4*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y1Conj
. Trans[Bar[y1Conj]]]) - (2*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3 + (76*Tr[y1 . Trans[Bar[y1]] . y1Conj .
Trans[Bar[y1Conj]]])/45 + (76*Tr[y1 . Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1]]])/45 - (2*Tr[y1Conj .
Trans[Bar[y1Conj]] . y1Conj . Trans[Bar[y1Conj]]])/3

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) -
24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2 -
(4*lambdaT3GeneralQ000211I2*Bar[lambdaT3GeneralQ000211I2])/3 + 4*lambdaS2Adj*Tr[y2 . Trans[Bar[y2]]] - (2*Tr[y2 .
Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambda12Adj =
-24*g2^2*lambda12Adj - 6*gY^2*lambda12Adj + 8*lambda12*lambda12Adj + 3*lambdaH1Adj*lambdaH2Adj + 2*lambda12Adj*lambdaS1
 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*lambda12Adj*lambdaS2Adj - (2*lambdaT3*Bar[lambdaT3])/9 -
(lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1])/3 + lambda12Adj*((16*lambda12Cross)/3 + 2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]]) + (2*Tr[y1 . Trans[Bar[y1]] . y2 .
Trans[Bar[y2]]])/3 - (2*Tr[y1Conj . Trans[Bar[y1Conj]] . y2 . Trans[Bar[y2]]])/3

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 6*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 +
2*lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
lambdaT3*Bar[lambdaT3] + 8*lambdaT3GeneralQ000211I1*Bar[lambdaT3GeneralQ000211I2] +
4*lambdaT3GeneralQ001300I1*Bar[lambdaT3GeneralQ000211I2] - lambdaT3GeneralQ021010I1*Bar[lambdaT3GeneralQ021010I1] +
lambdaT3GeneralQ000211I2*(8*Bar[lambdaT3GeneralQ000211I1] + 10*Bar[lambdaT3GeneralQ000211I2] +
4*Bar[lambdaT3GeneralQ001300I1] + 2*Tr[y1 . Trans[Bar[y1Conj]]]) + 2*Bar[lambdaT3GeneralQ000211I2]*Tr[y1Conj .
Trans[Bar[y1]]] + lambda12Cross*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 2*Tr[y2 .
Trans[Bar[y2]]]) - (4*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/5 - (4*Tr[y1Conj . Trans[Bar[y1Conj]] . y2 .
Trans[Bar[y2]]])/5

beta_y1Conj =
2*Tr[y1Conj . Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y1Conj][gen[$i], heavy[$j]])/2 -
(3*gY^2*Matrix[y1Conj][gen[$i], heavy[$j]])/2 + Tr[y1 . Trans[Bar[y1]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + Tr[y1Conj
. Trans[Bar[y1Conj]]]*Matrix[y1Conj][gen[$i], heavy[$j]] + (29*Matrix[y1, Trans[Bar[y1]], y1Conj][gen[$i], heavy[$j]] +
 29*Matrix[y1Conj, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 18*Matrix[y1Conj, Trans[Bar[y1Conj]], y1Conj][gen[$i],
heavy[$j]] + 9*Matrix[y2, Trans[Bar[y2]], y1Conj][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]], y1Conj][gen[$i],
heavy[$j]])/12

beta_lambdaT3GeneralQ000211I1 =
-24*g2^2*lambdaT3GeneralQ000211I1 - 6*gY^2*lambdaT3GeneralQ000211I1 + 2*lambdaS1*lambdaT3GeneralQ000211I1 +
lambdaS1Adj*(-4*lambdaT3GeneralQ000211I1 - 2*lambdaT3GeneralQ000211I2) + lambdaS2Adj*(4*lambdaT3GeneralQ000211I1 +
2*lambdaT3GeneralQ000211I2) + lambdaS2*(8*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2) +
lambda12Cross*(4*lambdaT3GeneralQ000211I2 + 4*lambdaT3GeneralQ001300I1) + 4*lambdaH2*lambdaT3GeneralQ110200I1 +
8*lambdaT3GeneralQ000400I1*(5*Bar[lambdaT3GeneralQ000211I1] + Bar[lambdaT3GeneralQ000211I2]) +
2*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] + lambda12*(8*lambdaT3GeneralQ000211I1 + 10*lambdaT3GeneralQ001300I1 +
2*Tr[y1Conj . Trans[Bar[y1]]]) + lambdaT3GeneralQ000211I1*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj .
Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (22*Tr[y1Conj . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/15

beta_lambdaT3GeneralQ000211I2 =
-24*g2^2*lambdaT3GeneralQ000211I2 - 6*gY^2*lambdaT3GeneralQ000211I2 + 8*lambda12*lambdaT3GeneralQ000211I2 +
2*lambdaS1*lambdaT3GeneralQ000211I2 + 2*lambdaS1Adj*lambdaT3GeneralQ000211I2 + 2*lambdaS2*lambdaT3GeneralQ000211I2 -
2*lambdaS2Adj*lambdaT3GeneralQ000211I2 + 16*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ000211I2] -
2*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] + lambda12Cross*(16*lambdaT3GeneralQ000211I1 +
20*lambdaT3GeneralQ000211I2 + 8*lambdaT3GeneralQ001300I1 + 4*Tr[y1Conj . Trans[Bar[y1]]]) +
lambdaT3GeneralQ000211I2*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 2*Tr[y2 . Trans[Bar[y2]]]) -
(8*Tr[y1Conj . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/5

beta_lambdaT3GeneralQ000400I1 =
3*lambdaT3GeneralQ000211I1^2 + 2*lambdaT3GeneralQ000211I1*lambdaT3GeneralQ000211I2 + lambdaT3GeneralQ000211I2^2 -
24*g2^2*lambdaT3GeneralQ000400I1 + 12*lambdaS1*lambdaT3GeneralQ000400I1 - 8*lambdaS1Adj*lambdaT3GeneralQ000400I1 +
11*lambdaT3GeneralQ001300I1^2 + 2*lambdaT3GeneralQ110200I1^2 + 2*lambdaT3GeneralQ001300I1*Tr[y1Conj . Trans[Bar[y1]]] +
 lambdaT3GeneralQ000400I1*(4*Tr[y1 . Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - (23*Tr[y1Conj .
```

```
Trans[Bar[y1]] . y1Conj . Trans[Bar[y1]]])/15

beta_lambdaT3GeneralQ001300I1 =
lambda12*(6*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2) + lambda12Cross*(4*lambdaT3GeneralQ000211I1 +
4*lambdaT3GeneralQ000211I2) - 24*g2^2*lambdaT3GeneralQ001300I1 + 4*lambdaH1*lambdaT3GeneralQ110200I1 +
lambdaT3GeneralQ000400I1*(88*Bar[lambdaT3GeneralQ001300I1] + 8*Tr[y1 . Trans[Bar[y1Conj]]]) +
lambdaS1*(22*lambdaT3GeneralQ001300I1 + 2*Tr[y1Conj . Trans[Bar[y1]]]) + lambdaT3GeneralQ001300I1*(4*Tr[y1 .
Trans[Bar[y1]]] + 4*Tr[y1Conj . Trans[Bar[y1Conj]]]) - (46*Tr[y1 . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1]]])/15 -
(46*Tr[y1Conj . Trans[Bar[y1]] . y1Conj . Trans[Bar[y1Conj]]])/15

beta_lambdaT3GeneralQ021010I1 =
(-33*g2^2*lambdaT3GeneralQ021010I1)/2 - (9*gY^2*lambdaT3GeneralQ021010I1)/2 + 2*lambda12*lambdaT3GeneralQ021010I1 -
2*lambda12Adj*lambdaT3GeneralQ021010I1 - 2*lambda12Cross*lambdaT3GeneralQ021010I1 + 2*lambdaH*lambdaT3GeneralQ021010I1
+ 4*lambdaH1*lambdaT3GeneralQ021010I1 - (4*lambdaH1Adj*lambdaT3GeneralQ021010I1)/3 +
4*lambdaH2*lambdaT3GeneralQ021010I1 - (4*lambdaH2Adj*lambdaT3GeneralQ021010I1)/3 +
2*Bar[lambdaT3]*(2*Bar[lambdaT3GeneralQ000211I1] - Bar[lambdaT3GeneralQ000211I2] + 4*Bar[lambdaT3GeneralQ110200I1]) +
2*Bar[lambdaT3]*Tr[y1 . Trans[Bar[y1Conj]]] + lambdaT3GeneralQ021010I1*(Tr[y1 . Trans[Bar[y1]]] + Tr[y1Conj .
Trans[Bar[y1Conj]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110200I1 =
lambdaH2*(6*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2) - (33*g2^2*lambdaT3GeneralQ110200I1)/2 -
(3*gY^2*lambdaT3GeneralQ110200I1)/2 + 6*lambdaH*lambdaT3GeneralQ110200I1 + 2*lambdaS1*lambdaT3GeneralQ110200I1 -
4*lambdaS1Adj*lambdaT3GeneralQ110200I1 + 4*Bar[lambdaT3]*Bar[lambdaT3GeneralQ021010I1] +
40*lambdaT3GeneralQ000400I1*Bar[lambdaT3GeneralQ110200I1] + lambdaH1*(10*lambdaT3GeneralQ001300I1 +
8*lambdaT3GeneralQ110200I1 + 2*Tr[y1Conj . Trans[Bar[y1]]]) + lambdaT3GeneralQ110200I1*(2*Tr[y1 . Trans[Bar[y1]]] +
2*Tr[y1Conj . Trans[Bar[y1Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
- 2*Tr[y1Conj . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_muT3GeneralS1Sq =
(6*lambdaT3GeneralQ000211I1 + 2*lambdaT3GeneralQ000211I2)*mS2Sq - 12*g2^2*muT3GeneralS1Sq + 2*lambdaS1*muT3GeneralS1Sq
- 4*lambdaS1Adj*muT3GeneralS1Sq + 40*lambdaT3GeneralQ000400I1*Bar[muT3GeneralS1Sq] + mS1Sq*(10*lambdaT3GeneralQ001300I1
 + 2*Tr[y1Conj . Trans[Bar[y1]]]) + muT3GeneralS1Sq*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y1Conj . Trans[Bar[y1Conj]]]) -
4*Tr[MF . Trans[Bar[MF]] . Trans[y1Conj] . Bar[y1]]
```

MODEL E_+2

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Extra quartics: 0; extra bilinears: 0; extra Yukawa: none
```

COMPLETE BSM LAGRANGIAN

```
-(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]) + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H, Scalar,
{Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG,
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1,
T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar1SelfCGInv2, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaS1Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 - Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {},
0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1,
SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
```

```
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv2,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv2, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3CrossScalarCGInv3, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambda12Inv3, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2,
 Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2,
T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv2, {}, 0]*Field[NewScalar2,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 - Coupling[MF, {},
0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*Coupling[y2, {Index[d$$1,
Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG,
{Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1,
NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
 Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1,
T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1,
 {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC,
Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(179*gY^3)/6

beta_g2 =
g2^3/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 2*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (6*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 6*Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (27*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*(Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]]))/4

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq + mS1Sq*(8*lambdaS1 + 4*lambdaS1Adj + 2*Tr[y1 .
```

```
Trans[Bar[y1]]]) - 4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq - 24*gY^2*mS2Sq + mS2Sq*(8*lambdaS2 + 4*lambdaS2Adj + 2*Tr[y2 .
Trans[Bar[y2]]]) - 4*Tr[MF . Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] +
4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 + gY^2*(8*g2^2 -
12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 16*lambdaS1Adj^2 + lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 .
Trans[Bar[y1]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_lambdaS2 =
22*g2^4 + 192*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 + gY^2*(32*g2^2 -
48*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 16*lambdaS2Adj^2 + lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 .
Trans[Bar[y2]]]) - (8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 +
(6*lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*lambdaS1Adj +
2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) -
2*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + 12*gY^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (51*gY^2*lambdaH2)/2 +
6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] +
lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
(63*g2^4)/5 + 48*gY^4 - 24*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 +
4*lambdaH1*lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 +
4*lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + 2*lambdaT3*Bar[lambdaT3] + lambda12*(4*lambdaS1Adj +
4*lambdaS2Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (22*Tr[y1 . Trans[Bar[y1]] . y2 .
Trans[Bar[y2]]])/15

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross - 2*lambdaH1Adj - (4*lambdaH2Adj)/3 + Tr[y1 .
Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
12*lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(16*lambdaS1Adj +
2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) -
(4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj + gY^2*(8*g2^2 - (51*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 + lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2
. Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (4*Tr[y2 .
Trans[Bar[y2]] . ye . Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) -
24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2 + 4*lambdaS1Adj*Tr[y1 . Trans[Bar[y1]]] - (2*Tr[y1 .
Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 + gY^2*(32*g2^2 - 48*lambdaS2Adj) -
24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2 + 4*lambdaS2Adj*Tr[y2 . Trans[Bar[y2]]] - (2*Tr[y2 .
Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambda12Adj =
gY^2*(16*g2^2 - 30*lambda12Adj) - 24*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + 3*lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*lambda12Adj*lambdaS2Adj -
(2*lambdaT3*Bar[lambdaT3])/9 + lambda12Adj*((16*lambda12Cross)/3 + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 .
Trans[Bar[y2]]]) + (2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 30*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 +
2*lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
lambdaT3*Bar[lambdaT3] + lambda12Cross*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (4*Tr[y1 .
Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/5
```

MODEL E_-2

Extra quartics: 6; extra bilinears: 1; extra Yukawa: y2Conj

COMPLETE BSM LAGRANGIAN

```
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3ExtraCGM000002I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraM000002I1, {}, 0] +
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000004I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ000004I1, {}, 0] + Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3ExtraCGQ020101I1, {Index[d$$1, SU2L[fund]],
Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraQ020101I1, {}, 0] +
Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3ExtraCGQ110002I1, {Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaT3ExtraQ110002I1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}] - Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS1, {},
0]^2*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{Index[d$$1, Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H,
Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
 T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG,
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3,
T3BSMd3]}, {}]]*CG[T3ExtraCGQ001102I1, {Index[d$$4, T3BSMd3], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ001102I1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]]*CG[T3ExtraCGQ001102I2, {Index[d$$4, T3BSMd3],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3]}]*Coupling[lambdaT3ExtraQ001102I2, {},
0]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv1, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv2,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv2, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 -
Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {}, 0]^2*Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + Bar[Coupling[lambdaT3, {},
0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1, SU2L[fund]], Index[d$$2,
SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
(Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
 T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Coupling[lambdaT3ExtraQ020101I1, {}, 0]]*CG[Bar[T3ExtraCGQ020101I1],
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[H,
Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] +
Bar[Coupling[lambdaT3ExtraM000002I1, {}, 0]]*CG[T3ExtraCGM000002I1, {Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]
+ Bar[Coupling[lambdaT3ExtraQ110002I1, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*CG[T3ExtraCGQ110002I1, {Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000013I1, {Index[d$$4, T3BSMd3], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3,
T3BSMd3]}]*Coupling[lambdaT3ExtraQ000013I1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
```

```
{}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2,
Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv2, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2,
T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv3, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv3, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + (Bar[Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv1,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv1, {},
0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 +
(Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar2SelfCGInv2, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambdaS2Inv2, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + Bar[Coupling[lambdaT3ExtraQ001102I1, {}, 0]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3ExtraCGQ001102I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3],
Index[d$$4, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$3,
T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ001102I2, {},
0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3ExtraCGQ001102I2, {Index[d$$1, T3BSMd3],
Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4, T3BSMd3]}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Coupling[lambdaT3ExtraQ000013I1, {}, 0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3ExtraCGQ000013I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Coupling[lambdaT3ExtraQ000004I1, {},
0]]*CG[T3ExtraCGQ000004I1, {Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] -
Coupling[MF, {}, 0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1,
SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {}]] + CG[T3Y2ConjCG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*Coupling[y2Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]*Field[NewScalar2, Scalar, {Index[d$$1,
T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[Proj[1]],
Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG, {Index[d$$1, SU2L[fund]],
Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1,
T3BSMd2], Index[d$$1, NFlavor]}, {}]]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2Conj, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3Y2ConjCG], {Bar[Index[d$$1, SU2L[fund]]],
Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] +
Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]],
Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], DiracProduct[Proj[-1]],
Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1, {Index[d$$1, Flavor],
Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1,
SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(59*gY^3)/6

beta_g2 =
g2^3/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
```

```
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + Matrix[y2Conj, Trans[Bar[y2Conj]], ye][gen[$i], gen[$j]] + 2*Matrix[ye,
Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (6*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2Conj, Trans[Bar[y2Conj]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye,
Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (3*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + Tr[y2Conj . Trans[Bar[y2Conj]]]*Matrix[y2][gen[$i], heavy[$j]] +
2*Tr[y2 . Trans[Bar[y2Conj]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + (9*Matrix[y1, Trans[Bar[y1]], y2][gen[$i],
heavy[$j]] + 18*Matrix[y2, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 29*Matrix[y2, Trans[Bar[y2Conj]], y2Conj][gen[$i],
 heavy[$j]] + 29*Matrix[y2Conj, Trans[Bar[y2Conj]], y2][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]], y2][gen[$i],
 heavy[$j]])/12

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (3*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*(Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + Matrix[MF, Trans[Bar[y2Conj]], y2Conj][heavy[$i], heavy[$j]] +
Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]]))/4

beta_mS1Sq =
-12*g2^2*mS1Sq - 6*gY^2*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq +
4*muT3GeneralS2Sq*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) +
12*lambdaT3GeneralQ001102I1*Bar[muT3GeneralS2Sq] + 4*lambdaT3GeneralQ001102I2*Bar[muT3GeneralS2Sq] + mS1Sq*(8*lambdaS1
+ 4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]]) - 4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq + 20*lambdaT3GeneralQ000013I1*Bar[muT3GeneralS2Sq] +
4*Bar[muT3GeneralS2Sq]*Tr[y2 . Trans[Bar[y2Conj]]] + muT3GeneralS2Sq*(20*Bar[lambdaT3GeneralQ000013I1] + 4*Tr[y2Conj .
Trans[Bar[y2]]]) + mS2Sq*(8*lambdaS2 + 4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
 4*Tr[MF . Trans[Bar[y2]] . y2 . Trans[Bar[MF]]] - 4*Tr[MF . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 24*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1] +
lambdaH*(12*Tr[yd . Trans[Bar[yd]]] + 4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd .
Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye . Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] .
yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 + gY^2*(8*g2^2 -
12*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 16*lambdaS1Adj^2 +
8*lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) +
lambdaT3GeneralQ001102I2*(8*Bar[lambdaT3GeneralQ001102I1] + (20*Bar[lambdaT3GeneralQ001102I2])/3) +
lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 . Trans[Bar[y1]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_lambdaS2 =
22*g2^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 - 24*g2^2*lambdaS2 +
14*lambdaS2^2 + 16*lambdaS2Adj^2 + (832*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 +
(8*lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]))/3 +
lambdaT3GeneralQ001102I2*((8*Bar[lambdaT3GeneralQ001102I1])/3 + 4*Bar[lambdaT3GeneralQ001102I2]) +
(16*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3 + (32*Bar[lambdaT3GeneralQ000013I1]*Tr[y2 .
Trans[Bar[y2Conj]]])/3 + lambdaT3GeneralQ000013I1*((352*Bar[lambdaT3GeneralQ000013I1])/3 + (32*Tr[y2Conj .
Trans[Bar[y2]]])/3) + lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
(8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3 - (248*Tr[y2 . Trans[Bar[y2]] . y2Conj . Trans[Bar[y2Conj]]])/45 -
(248*Tr[y2 . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2]]])/45 - (8*Tr[y2Conj . Trans[Bar[y2Conj]] . y2Conj .
Trans[Bar[y2Conj]]])/3

beta_lambdaH1 =
6*g2^4 + 3*gY^4 - (33*g2^2*lambdaH1)/2 - (15*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 +
(6*lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ110002I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) +
4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 12*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ110002I1] +
4*lambdaT3GeneralQ001102I2*Bar[lambdaT3GeneralQ110002I1] + lambdaH1*(4*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]] +
6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y1 . Trans[Bar[y1]] . ye .
Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (3*gY^2*lambdaH2)/2 + 6*lambdaH*lambdaH2 +
4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 20*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ110002I1] +
4*Bar[lambdaT3GeneralQ110002I1]*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ110002I1*(4*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ110002I1]) + 4*Tr[y2Conj .
Trans[Bar[y2]]]) + lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd .
Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye .
Trans[Bar[ye]]] - 2*Tr[y2Conj . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
```

```
(63*g2^4)/5 - 24*g2^2*lambda12 - 6*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 +
4*lambdaH1*lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 +
4*lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + 2*lambdaT3*Bar[lambdaT3] +
4*lambdaT3GeneralQ001102I2*(Bar[lambdaT3GeneralQ000013I1] + Bar[lambdaT3GeneralQ001102I2]) +
4*lambdaT3GeneralQ000013I1*(5*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]) +
2*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] + 4*Bar[lambdaT3GeneralQ001102I1]*Tr[y2 . Trans[Bar[y2Conj]]]
+ lambdaT3GeneralQ001102I1*(4*(5*Bar[lambdaT3GeneralQ000013I1] + 4*Bar[lambdaT3GeneralQ001102I1]) + 4*Tr[y2Conj .
Trans[Bar[y2]]]) + lambda12*(4*lambdaS1Adj + 4*lambdaS2Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] +
2*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (22*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/15 - (22*Tr[y1 . Trans[Bar[y1]]
 . y2Conj . Trans[Bar[y2Conj]]])/15

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (9*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ020101I1] -
2*lambdaT3GeneralQ001102I2*Bar[lambdaT3GeneralQ020101I1] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ020101I1] +
2*Bar[lambdaT3GeneralQ020101I1]*Tr[y2 . Trans[Bar[y2Conj]]] + lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross -
2*lambdaH1Adj - (4*lambdaH2Adj)/3 + Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + Tr[y2Conj . Trans[Bar[y2Conj]]]
 + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-4*g2^2 - (15*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
12*lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 -
(4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1])/3 + lambdaH1Adj*(16*lambdaS1Adj + 2*Tr[y1 . Trans[Bar[y1]]]
+ 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) - (4*Tr[y1 . Trans[Bar[y1]] . ye .
 Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj - (33*g2^2*lambdaH2Adj)/2 - (3*gY^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 -
(4*lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1])/3 + lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]]
+ 2*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]]) + (4*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]])/3 - (4*Tr[y2Conj . Trans[Bar[y2Conj]] . ye .
Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 + gY^2*(8*g2^2 - 12*lambdaS1Adj) -
24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2 -
(4*lambdaT3GeneralQ001102I2*Bar[lambdaT3GeneralQ001102I2])/3 + 4*lambdaS1Adj*Tr[y1 . Trans[Bar[y1]]] - (2*Tr[y1 .
Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 - 24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj +
20*lambdaS2Adj^2 - (224*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ000004I1])/3 -
(4*lambdaT3GeneralQ001102I2*Bar[lambdaT3GeneralQ001102I1])/3 -
(4*lambdaT3GeneralQ001102I1*(3*Bar[lambdaT3GeneralQ001102I1] + Bar[lambdaT3GeneralQ001102I2]))/3 -
(8*lambdaT3GeneralQ110002I1*Bar[lambdaT3GeneralQ110002I1])/3 - (4*Bar[lambdaT3GeneralQ000013I1]*Tr[y2 .
Trans[Bar[y2Conj]]])/3 + lambdaT3GeneralQ000013I1*((-44*Bar[lambdaT3GeneralQ000013I1])/3 - (4*Tr[y2Conj .
Trans[Bar[y2]]])/3) + lambdaS2Adj*(4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (2*Tr[y2 .
Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3 + (76*Tr[y2 . Trans[Bar[y2]] . y2Conj . Trans[Bar[y2Conj]]])/45 + (76*Tr[y2 .
Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2]]])/45 - (2*Tr[y2Conj . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2Conj]]])/3

beta_lambda12Adj =
-24*g2^2*lambda12Adj - 6*gY^2*lambda12Adj + 8*lambda12*lambda12Adj + 3*lambdaH1Adj*lambdaH2Adj + 2*lambda12Adj*lambdaS1
 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*lambda12Adj*lambdaS2Adj - (2*lambdaT3*Bar[lambdaT3])/9 -
(lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1])/3 + lambda12Adj*((16*lambda12Cross)/3 + 2*Tr[y1 .
Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) + (2*Tr[y1 . Trans[Bar[y1]] . y2 .
Trans[Bar[y2]]])/3 - (2*Tr[y1 . Trans[Bar[y1]] . y2Conj . Trans[Bar[y2Conj]]])/3

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 6*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 +
2*lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
lambdaT3*Bar[lambdaT3] + 4*lambdaT3GeneralQ000013I1*Bar[lambdaT3GeneralQ001102I2] +
8*lambdaT3GeneralQ001102I1*Bar[lambdaT3GeneralQ001102I2] - lambdaT3GeneralQ020101I1*Bar[lambdaT3GeneralQ020101I1] +
2*Bar[lambdaT3GeneralQ001102I2]*Tr[y2 . Trans[Bar[y2Conj]]] + lambdaT3GeneralQ001102I2*(4*Bar[lambdaT3GeneralQ000013I1]
 + 8*Bar[lambdaT3GeneralQ001102I1] + 10*Bar[lambdaT3GeneralQ001102I2] + 2*Tr[y2Conj . Trans[Bar[y2]]]) +
lambda12Cross*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (4*Tr[y1 .
Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/5 - (4*Tr[y1 . Trans[Bar[y1]] . y2Conj . Trans[Bar[y2Conj]]])/5

beta_y2Conj =
2*Tr[y2Conj . Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] - (9*g2^2*Matrix[y2Conj][gen[$i], heavy[$j]])/2 -
(3*gY^2*Matrix[y2Conj][gen[$i], heavy[$j]])/2 + Tr[y2 . Trans[Bar[y2]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + Tr[y2Conj
. Trans[Bar[y2Conj]]]*Matrix[y2Conj][gen[$i], heavy[$j]] + (9*Matrix[y1, Trans[Bar[y1]], y2Conj][gen[$i], heavy[$j]] +
29*Matrix[y2, Trans[Bar[y2]], y2Conj][gen[$i], heavy[$j]] + 29*Matrix[y2Conj, Trans[Bar[y2]], y2][gen[$i], heavy[$j]] +
 18*Matrix[y2Conj, Trans[Bar[y2Conj]], y2Conj][gen[$i], heavy[$j]] + 6*Matrix[ye, Trans[Bar[ye]], y2Conj][gen[$i],
heavy[$j]])/12

beta_lambdaT3GeneralQ000004I1 =
-24*g2^2*lambdaT3GeneralQ000004I1 + 12*lambdaS2*lambdaT3GeneralQ000004I1 - 8*lambdaS2Adj*lambdaT3GeneralQ000004I1 +
11*lambdaT3GeneralQ000013I1^2 + 3*lambdaT3GeneralQ001102I1^2 + 2*lambdaT3GeneralQ001102I1*lambdaT3GeneralQ001102I2 +
lambdaT3GeneralQ001102I2^2 + 2*lambdaT3GeneralQ110002I1^2 + 2*lambdaT3GeneralQ000013I1*Tr[y2 . Trans[Bar[y2Conj]]] +
lambdaT3GeneralQ000004I1*(4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (23*Tr[y2 .
```

```
Trans[Bar[y2Conj]] . y2 . Trans[Bar[y2Conj]]])/15

beta_lambdaT3GeneralQ000013I1 =
-24*g2^2*lambdaT3GeneralQ000013I1 + lambda12*(6*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2) +
lambda12Cross*(4*lambdaT3GeneralQ001102I1 + 4*lambdaT3GeneralQ001102I2) + 4*lambdaH2*lambdaT3GeneralQ110002I1 +
lambdaS2*(22*lambdaT3GeneralQ000013I1 + 2*Tr[y2 . Trans[Bar[y2Conj]]]) +
lambdaT3GeneralQ000004I1*(88*Bar[lambdaT3GeneralQ000013I1] + 8*Tr[y2Conj . Trans[Bar[y2]]]) +
lambdaT3GeneralQ000013I1*(4*Tr[y2 . Trans[Bar[y2]]] + 4*Tr[y2Conj . Trans[Bar[y2Conj]]]) - (46*Tr[y2 . Trans[Bar[y2]] .
 y2 . Trans[Bar[y2Conj]]])/15 - (46*Tr[y2 . Trans[Bar[y2Conj]] . y2Conj . Trans[Bar[y2Conj]]])/15

beta_lambdaT3GeneralQ001102I1 =
-24*g2^2*lambdaT3GeneralQ001102I1 - 6*gY^2*lambdaT3GeneralQ001102I1 + 2*lambdaS2*lambdaT3GeneralQ001102I1 +
lambdaS2Adj*(-4*lambdaT3GeneralQ001102I1 - 2*lambdaT3GeneralQ001102I2) + lambdaS1Adj*(4*lambdaT3GeneralQ001102I1 +
2*lambdaT3GeneralQ001102I2) + lambdaS1*(8*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2) +
lambda12Cross*(4*lambdaT3GeneralQ000013I1 + 4*lambdaT3GeneralQ001102I2) + 2*lambdaT3*lambdaT3GeneralQ020101I1 +
4*lambdaH1*lambdaT3GeneralQ110002I1 + 8*lambdaT3GeneralQ000004I1*(5*Bar[lambdaT3GeneralQ001102I1] +
Bar[lambdaT3GeneralQ001102I2]) + lambda12*(10*lambdaT3GeneralQ000013I1 + 8*lambdaT3GeneralQ001102I1 + 2*Tr[y2 .
Trans[Bar[y2Conj]]]) + lambdaT3GeneralQ001102I1*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj .
Trans[Bar[y2Conj]]]) - (22*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2Conj]]])/15

beta_lambdaT3GeneralQ001102I2 =
-24*g2^2*lambdaT3GeneralQ001102I2 - 6*gY^2*lambdaT3GeneralQ001102I2 + 8*lambda12*lambdaT3GeneralQ001102I2 +
2*lambdaS1*lambdaT3GeneralQ001102I2 - 2*lambdaS1Adj*lambdaT3GeneralQ001102I2 + 2*lambdaS2*lambdaT3GeneralQ001102I2 +
2*lambdaS2Adj*lambdaT3GeneralQ001102I2 - 2*lambdaT3*lambdaT3GeneralQ020101I1 +
16*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ001102I2] + lambda12Cross*(8*lambdaT3GeneralQ000013I1 +
16*lambdaT3GeneralQ001102I1 + 20*lambdaT3GeneralQ001102I2 + 4*Tr[y2 . Trans[Bar[y2Conj]]]) +
lambdaT3GeneralQ001102I2*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
(8*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2Conj]]])/5

beta_lambdaT3GeneralQ020101I1 =
(-33*g2^2*lambdaT3GeneralQ020101I1)/2 - (9*gY^2*lambdaT3GeneralQ020101I1)/2 + 2*lambda12*lambdaT3GeneralQ020101I1 -
2*lambda12Adj*lambdaT3GeneralQ020101I1 - 2*lambda12Cross*lambdaT3GeneralQ020101I1 + 2*lambdaH*lambdaT3GeneralQ020101I1
+ 4*lambdaH1*lambdaT3GeneralQ020101I1 - 2*lambdaH1Adj*lambdaT3GeneralQ020101I1 + 4*lambdaH2*lambdaT3GeneralQ020101I1 -
2*lambdaH2Adj*lambdaT3GeneralQ020101I1 + 4*lambdaT3GeneralQ001102I1*Bar[lambdaT3] -
2*lambdaT3GeneralQ001102I2*Bar[lambdaT3] + 8*lambdaT3GeneralQ110002I1*Bar[lambdaT3] + 2*Bar[lambdaT3]*Tr[y2 .
Trans[Bar[y2Conj]]] + lambdaT3GeneralQ020101I1*(Tr[y1 . Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + Tr[y2Conj .
Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])

beta_lambdaT3GeneralQ110002I1 =
lambdaH1*(6*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2) + 4*lambdaT3*lambdaT3GeneralQ020101I1 -
(33*g2^2*lambdaT3GeneralQ110002I1)/2 - (3*gY^2*lambdaT3GeneralQ110002I1)/2 + 6*lambdaH*lambdaT3GeneralQ110002I1 +
2*lambdaS2*lambdaT3GeneralQ110002I1 - 4*lambdaS2Adj*lambdaT3GeneralQ110002I1 +
40*lambdaT3GeneralQ000004I1*Bar[lambdaT3GeneralQ110002I1] + lambdaH2*(10*lambdaT3GeneralQ000013I1 +
8*lambdaT3GeneralQ110002I1 + 2*Tr[y2 . Trans[Bar[y2Conj]]]) + lambdaT3GeneralQ110002I1*(2*Tr[y2 . Trans[Bar[y2]]] +
2*Tr[y2Conj . Trans[Bar[y2Conj]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]])
- 2*Tr[y2 . Trans[Bar[y2Conj]] . ye . Trans[Bar[ye]]]

beta_muT3GeneralS2Sq =
(6*lambdaT3GeneralQ001102I1 + 2*lambdaT3GeneralQ001102I2)*mS1Sq - 12*g2^2*muT3GeneralS2Sq + 2*lambdaS2*muT3GeneralS2Sq
- 4*lambdaS2Adj*muT3GeneralS2Sq + 40*lambdaT3GeneralQ000004I1*Bar[muT3GeneralS2Sq] + mS2Sq*(10*lambdaT3GeneralQ000013I1
 + 2*Tr[y2 . Trans[Bar[y2Conj]]]) + muT3GeneralS2Sq*(2*Tr[y2 . Trans[Bar[y2]]] + 2*Tr[y2Conj . Trans[Bar[y2Conj]]]) -
4*Tr[MF . Trans[Bar[y2Conj]] . y2 . Trans[Bar[MF]]]
```

MODEL E_-4

Extra quartics: 0; extra bilinears: 0; extra Yukawa: none

COMPLETE BSM LAGRANGIAN

```
-(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS1, {}, 0]^2*Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]) + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1,
Lorentz]}]]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] + (Bar[Field[H, Scalar,
{Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH1Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal1CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaH1Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar1, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3MixCG,
{Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1,
T3BSMd3]}]*Coupling[lambdaT3, {}, 0]*Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]*Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]*Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}] + (Bar[Field[NewScalar1, Scalar, {Index[d$$1,
T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar1SelfCGInv1, {Index[d$$1,
T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS1Inv1, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 +
(Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3Scalar1SelfCGInv2, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
```

```
T3BSMd3]}]*Coupling[lambdaS1Inv2, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar1,
Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 - Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Coupling[MS2, {},
0]^2*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{Index[d$$1, Lorentz]}]]*Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {Index[d$$1, Lorentz]}] +
Bar[Coupling[lambdaT3, {}, 0]]*Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[H, Scalar, {Index[d$$2,
SU2L[fund]]}, {}]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[Bar[T3MixCG], {Index[d$$1,
SU2L[fund]], Index[d$$2, SU2L[fund]], Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}] + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$1, T3BSMd3], Index[d$$2, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv1, {Index[d$$1, SU2L[fund]],
Bar[Index[d$$2, SU2L[fund]]], Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv1, {}, 0]*Field[H,
Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar,
 {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2,
{Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]], Index[d$$1, T3BSMd3], Index[d$$2,
T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]}, {}]*Field[NewScalar2, Scalar,
{Index[d$$2, T3BSMd3]}, {}])/2 + (Bar[Field[H, Scalar, {Index[d$$1, SU2L[fund]]}, {}]]*Bar[Field[NewScalar2, Scalar,
{Index[d$$1, T3BSMd3]}, {}]]*CG[T3HiggsPortal2CGInv2, {Index[d$$1, SU2L[fund]], Bar[Index[d$$2, SU2L[fund]]],
Index[d$$2, T3BSMd3], Index[d$$1, T3BSMd3]}]*Coupling[lambdaH2Inv2, {}, 0]*Field[H, Scalar, {Index[d$$2, SU2L[fund]]},
{}]*Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}])/2 + Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv1, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv1, {}, 0]*Field[NewScalar1,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] + Bar[Field[NewScalar1,
Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3CrossScalarCGInv2,
 {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambda12Inv2, {},
0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}] +
Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]},
{}]]*CG[T3CrossScalarCGInv3, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4,
T3BSMd3]}]*Coupling[lambda12Inv3, {}, 0]*Field[NewScalar1, Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2,
Scalar, {Index[d$$4, T3BSMd3]}, {}] + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]}, {}]]*Bar[Field[NewScalar2,
 Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv1, {Index[d$$1, T3BSMd3], Index[d$$3, T3BSMd3], Index[d$$2,
T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv1, {}, 0]*Field[NewScalar2, Scalar, {Index[d$$3, T3BSMd3]},
{}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 + (Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*Bar[Field[NewScalar2, Scalar, {Index[d$$2, T3BSMd3]}, {}]]*CG[T3Scalar2SelfCGInv2, {Index[d$$1, T3BSMd3],
Index[d$$3, T3BSMd3], Index[d$$2, T3BSMd3], Index[d$$4, T3BSMd3]}]*Coupling[lambdaS2Inv2, {}, 0]*Field[NewScalar2,
Scalar, {Index[d$$3, T3BSMd3]}, {}]*Field[NewScalar2, Scalar, {Index[d$$4, T3BSMd3]}, {}])/2 - Coupling[MF, {},
0]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]], Field[NewFermion, Fermion,
{Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + Bar[Field[NewScalar2, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[T3Y2CG, {Index[d$$1, SU2L[fund]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1, T3BSMd3]}]*Coupling[y2, {Index[d$$1,
Flavor], Index[d$$1, NFlavor]}, 0]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]],
DiracProduct[Proj[1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]] + CG[T3Y1CG,
{Index[d$$1, SU2L[fund]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Coupling[y1, {Index[d$$1, Flavor], Index[d$$1,
NFlavor]}, 0]*Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[l, Fermion, {Index[d$$1, SU2L[fund]],
 Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC, Proj[1]], Transp[Bar[Field[NewFermion, Fermion, {Index[d$$1,
T3BSMd2], Index[d$$1, NFlavor]}, {}]]] + I*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1,
NFlavor]}, {}]], DiracProduct[GammaM[Index[d$$1, Lorentz]]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2],
Index[d$$1, NFlavor]}, {Index[d$$1, Lorentz]}]] + Bar[Coupling[y2, {Index[d$$1, Flavor], Index[d$$1, NFlavor]},
0]]*CG[Bar[T3Y2CG], {Bar[Index[d$$1, SU2L[fund]]], Index[d$$1, T3BSMd2], Index[d$$1, T3BSMd3]}]*Field[NewScalar2,
Scalar, {Index[d$$1, T3BSMd3]}, {}]*NCM[Bar[Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]},
{}]], DiracProduct[Proj[-1]], Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]] + Bar[Coupling[y1,
 {Index[d$$1, Flavor], Index[d$$1, NFlavor]}, 0]]*Bar[Field[NewScalar1, Scalar, {Index[d$$1, T3BSMd3]},
{}]]*CG[Bar[T3Y1CG], {Bar[Index[d$$1, SU2L[fund]]], Bar[Index[d$$1, T3BSMd2]], Index[d$$1,
T3BSMd3]}]*NCM[Transp[Field[l, Fermion, {Index[d$$1, SU2L[fund]], Index[d$$1, Flavor]}, {}]], DiracProduct[GammaCC,
Proj[-1]], Field[NewFermion, Fermion, {Index[d$$1, T3BSMd2], Index[d$$1, NFlavor]}, {}]]
```

UV RENORMALISABLE BETA FUNCTIONS

```
beta_gY =
(179*gY^3)/6

beta_g2 =
g2^3/6

beta_g3 =
-7*g3^3

beta_yu =
(-9*g2^2*Matrix[yu][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yu][gen[$i], gen[$j]] - (17*gY^2*Matrix[yu][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yu][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yu][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yu][gen[$i], gen[$j]] - (3*(Matrix[yd, Trans[Bar[yd]], yu][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yu][gen[$i], gen[$j]]))/2

beta_yd =
(-9*g2^2*Matrix[yd][gen[$i], gen[$j]])/4 - 8*g3^2*Matrix[yd][gen[$i], gen[$j]] - (5*gY^2*Matrix[yd][gen[$i],
gen[$j]])/12 + 3*Tr[yd . Trans[Bar[yd]]]*Matrix[yd][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[yd][gen[$i],
gen[$j]] + 3*Tr[yu . Trans[Bar[yu]]]*Matrix[yd][gen[$i], gen[$j]] + (3*(Matrix[yd, Trans[Bar[yd]], yd][gen[$i],
gen[$j]] - Matrix[yu, Trans[Bar[yu]], yd][gen[$i], gen[$j]]))/2

beta_ye =
(-9*g2^2*Matrix[ye][gen[$i], gen[$j]])/4 - (15*gY^2*Matrix[ye][gen[$i], gen[$j]])/4 + 3*Tr[yd .
Trans[Bar[yd]]]*Matrix[ye][gen[$i], gen[$j]] + Tr[ye . Trans[Bar[ye]]]*Matrix[ye][gen[$i], gen[$j]] + 3*Tr[yu .
Trans[Bar[yu]]]*Matrix[ye][gen[$i], gen[$j]] + (3*(Matrix[y1, Trans[Bar[y1]], ye][gen[$i], gen[$j]] + Matrix[y2,
```

```
Trans[Bar[y2]], ye][gen[$i], gen[$j]] + 2*Matrix[ye, Trans[Bar[ye]], ye][gen[$i], gen[$j]]))/4

beta_y1 =
(-9*g2^2*Matrix[y1][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y1][gen[$i], heavy[$j]])/2 + Tr[y1 .
Trans[Bar[y1]]]*Matrix[y1][gen[$i], heavy[$j]] + (6*Matrix[y1, Trans[Bar[y1]], y1][gen[$i], heavy[$j]] + 3*Matrix[y2,
Trans[Bar[y2]], y1][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y1][gen[$i], heavy[$j]])/4

beta_y2 =
(-9*g2^2*Matrix[y2][gen[$i], heavy[$j]])/2 - (15*gY^2*Matrix[y2][gen[$i], heavy[$j]])/2 + Tr[y2 .
Trans[Bar[y2]]]*Matrix[y2][gen[$i], heavy[$j]] + (3*Matrix[y1, Trans[Bar[y1]], y2][gen[$i], heavy[$j]] + 6*Matrix[y2,
Trans[Bar[y2]], y2][gen[$i], heavy[$j]] + 2*Matrix[ye, Trans[Bar[ye]], y2][gen[$i], heavy[$j]])/4

beta_MF =
(-9*g2^2*Matrix[MF][heavy[$i], heavy[$j]])/2 - (27*gY^2*Matrix[MF][heavy[$i], heavy[$j]])/2 + (3*(Matrix[MF,
Trans[Bar[y2]], y2][heavy[$i], heavy[$j]] + Matrix[Trans[y1], Bar[y1], MF][heavy[$i], heavy[$j]]))/4

beta_mS1Sq =
-12*g2^2*mS1Sq - 24*gY^2*mS1Sq + (6*lambda12 + 4*lambda12Cross)*mS2Sq + mS1Sq*(8*lambdaS1 + 4*lambdaS1Adj + 2*Tr[y1 .
Trans[Bar[y1]]]) - 4*Tr[MF . Trans[Bar[MF]] . Trans[y1] . Bar[y1]]

beta_mS2Sq =
(6*lambda12 + 4*lambda12Cross)*mS1Sq - 12*g2^2*mS2Sq - 6*gY^2*mS2Sq + mS2Sq*(8*lambdaS2 + 4*lambdaS2Adj + 2*Tr[y2 .
Trans[Bar[y2]]]) - 4*Tr[MF . Trans[Bar[y2]] . y2 . Trans[Bar[MF]]]

beta_lambdaH =
(9*g2^4)/4 + (3*gY^4)/4 + gY^2*((3*g2^2)/2 - 3*lambdaH) - 9*g2^2*lambdaH + 12*lambdaH^2 + 6*lambdaH1^2 +
3*lambdaH1Adj^2 + 6*lambdaH2^2 + 3*lambdaH2Adj^2 + 4*lambdaT3*Bar[lambdaT3] + lambdaH*(12*Tr[yd . Trans[Bar[yd]]] +
4*Tr[ye . Trans[Bar[ye]]] + 12*Tr[yu . Trans[Bar[yu]]]) - 12*Tr[yd . Trans[Bar[yd]] . yd . Trans[Bar[yd]]] - 4*Tr[ye .
Trans[Bar[ye]] . ye . Trans[Bar[ye]]] - 12*Tr[yu . Trans[Bar[yu]] . yu . Trans[Bar[yu]]]

beta_lambdaS1 =
22*g2^4 + 192*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH1^2 + gY^2*(32*g2^2 -
48*lambdaS1) - 24*g2^2*lambdaS1 + 14*lambdaS1^2 + 16*lambdaS1Adj^2 + lambdaS1*(8*lambdaS1Adj + 4*Tr[y1 .
Trans[Bar[y1]]]) - (8*Tr[y1 . Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3

beta_lambdaS2 =
22*g2^4 + 12*gY^4 + 6*lambda12^2 + 8*lambda12*lambda12Cross + (20*lambda12Cross^2)/3 + 4*lambdaH2^2 + gY^2*(8*g2^2 -
12*lambdaS2) - 24*g2^2*lambdaS2 + 14*lambdaS2^2 + 16*lambdaS2Adj^2 + lambdaS2*(8*lambdaS2Adj + 4*Tr[y2 .
Trans[Bar[y2]]]) - (8*Tr[y2 . Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambdaH1 =
6*g2^4 + 12*gY^4 - (33*g2^2*lambdaH1)/2 - (51*gY^2*lambdaH1)/2 + 6*lambdaH*lambdaH1 + 4*lambdaH1^2 + 6*lambdaH1Adj^2 +
(6*lambda12 + 4*lambda12Cross)*lambdaH2 + 8*lambdaH1*lambdaS1 + 4*lambdaT3*Bar[lambdaT3] + lambdaH1*(4*lambdaS1Adj +
2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) -
2*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]]

beta_lambdaH2 =
6*g2^4 + 3*gY^4 + (6*lambda12 + 4*lambda12Cross)*lambdaH1 - (33*g2^2*lambdaH2)/2 - (15*gY^2*lambdaH2)/2 +
6*lambdaH*lambdaH2 + 4*lambdaH2^2 + 6*lambdaH2Adj^2 + 8*lambdaH2*lambdaS2 + 4*lambdaT3*Bar[lambdaT3] +
lambdaH2*(4*lambdaS2Adj + 2*Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
 Trans[Bar[yu]]]) - 2*Tr[y2 . Trans[Bar[y2]] . ye . Trans[Bar[ye]]]

beta_lambda12 =
(63*g2^4)/5 + 48*gY^4 - 24*g2^2*lambda12 - 30*gY^2*lambda12 + 4*lambda12^2 + 16*lambda12Adj^2 + 4*lambda12Cross^2 +
4*lambdaH1*lambdaH2 + (8*lambda12 + 4*lambda12Cross)*lambdaS1 + 4*lambda12Cross*lambdaS1Adj + (8*lambda12 +
4*lambda12Cross)*lambdaS2 + 4*lambda12Cross*lambdaS2Adj + 2*lambdaT3*Bar[lambdaT3] + lambda12*(4*lambdaS1Adj +
4*lambdaS2Adj + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (22*Tr[y1 . Trans[Bar[y1]] . y2 .
Trans[Bar[y2]]])/15

beta_lambdaT3 =
(-33*g2^2*lambdaT3)/2 - (33*gY^2*lambdaT3)/2 + 2*lambda12*lambdaT3 + 2*lambdaH*lambdaT3 + 4*lambdaH1*lambdaT3 +
4*lambdaH2*lambdaT3 + lambdaT3*((-4*lambda12Adj)/3 - 2*lambda12Cross - 2*lambdaH1Adj - (4*lambdaH2Adj)/3 + Tr[y1 .
Trans[Bar[y1]]] + Tr[y2 . Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu .
Trans[Bar[yu]]])

beta_lambdaH1Adj =
gY^2*(-8*g2^2 - (51*lambdaH1Adj)/2) - (33*g2^2*lambdaH1Adj)/2 + 2*lambdaH*lambdaH1Adj + 8*lambdaH1*lambdaH1Adj +
12*lambda12Adj*lambdaH2Adj + 2*lambdaH1Adj*lambdaS1 - (4*lambdaT3*Bar[lambdaT3])/3 + lambdaH1Adj*(16*lambdaS1Adj +
2*Tr[y1 . Trans[Bar[y1]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) -
(4*Tr[y1 . Trans[Bar[y1]] . ye . Trans[Bar[ye]]])/3

beta_lambdaH2Adj =
12*lambda12Adj*lambdaH1Adj + gY^2*(-4*g2^2 - (15*lambdaH2Adj)/2) - (33*g2^2*lambdaH2Adj)/2 + 2*lambdaH*lambdaH2Adj +
8*lambdaH2*lambdaH2Adj + 2*lambdaH2Adj*lambdaS2 - (8*lambdaT3*Bar[lambdaT3])/9 + lambdaH2Adj*(16*lambdaS2Adj + 2*Tr[y2
. Trans[Bar[y2]]] + 6*Tr[yd . Trans[Bar[yd]]] + 2*Tr[ye . Trans[Bar[ye]]] + 6*Tr[yu . Trans[Bar[yu]]]) + (4*Tr[y2 .
Trans[Bar[y2]] . ye . Trans[Bar[ye]]])/3

beta_lambdaS1Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH1Adj^2 + gY^2*(32*g2^2 - 48*lambdaS1Adj) -
24*g2^2*lambdaS1Adj + 12*lambdaS1*lambdaS1Adj + 20*lambdaS1Adj^2 + 4*lambdaS1Adj*Tr[y1 . Trans[Bar[y1]]] - (2*Tr[y1 .
Trans[Bar[y1]] . y1 . Trans[Bar[y1]]])/3
```

```
beta_lambdaS2Adj =
-2*g2^4 + 12*lambda12Adj^2 - (4*lambda12Cross^2)/3 + 3*lambdaH2Adj^2 + gY^2*(8*g2^2 - 12*lambdaS2Adj) -
24*g2^2*lambdaS2Adj + 12*lambdaS2*lambdaS2Adj + 20*lambdaS2Adj^2 + 4*lambdaS2Adj*Tr[y2 . Trans[Bar[y2]]] - (2*Tr[y2 .
Trans[Bar[y2]] . y2 . Trans[Bar[y2]]])/3

beta_lambda12Adj =
gY^2*(16*g2^2 - 30*lambda12Adj) - 24*g2^2*lambda12Adj + 8*lambda12*lambda12Adj + 3*lambdaH1Adj*lambdaH2Adj +
2*lambda12Adj*lambdaS1 + 16*lambda12Adj*lambdaS1Adj + 2*lambda12Adj*lambdaS2 + 16*lambda12Adj*lambdaS2Adj -
(2*lambdaT3*Bar[lambdaT3])/9 + lambda12Adj*((16*lambda12Cross)/3 + 2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 .
Trans[Bar[y2]]]) + (2*Tr[y1 . Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/3

beta_lambda12Cross =
(51*g2^4)/10 - 24*g2^2*lambda12Cross - 30*gY^2*lambda12Cross + 8*lambda12*lambda12Cross + 10*lambda12Cross^2 +
2*lambda12Cross*lambdaS1 - 2*lambda12Cross*lambdaS1Adj + 2*lambda12Cross*lambdaS2 - 2*lambda12Cross*lambdaS2Adj -
lambdaT3*Bar[lambdaT3] + lambda12Cross*(2*Tr[y1 . Trans[Bar[y1]]] + 2*Tr[y2 . Trans[Bar[y2]]]) - (4*Tr[y1 .
Trans[Bar[y1]] . y2 . Trans[Bar[y2]]])/5
```